Breaking Frontier Lock: Score Dilution, Certificate Gating, and Online Operator Induction

Yan Su

Abstract

Neuro-symbolic world-model synthesis solves 183/183 Public-25 levels at $G = 99.58$ (100.00 loop-erased, 0.413x human, 0tok) and 1,793/1,794 levels across 316 games, yet cutoffs and open-loop probing backfire across 825 baseline, 250 confirmation, and 30 Kaggle runs. Agents spend 52.4-82.9% of actions on their final unsolved level (≤ 0.091 hazard at 80), where Kaplan-Meier matching attributes 65-100% of slowdown to walls: 80-action cutoffs lose 16 clears and 14K truncation drops Hidden-110 from 24.62±2.22 (28.38) to 9.39 LB. Under $(b_l/a_l)^2$, Chebyshev's identity proves failed k-step probes dilute score after RESET, while 0-parameter geometric ($R^2 = 0.978, 0.997$) and sparse (MAE = 0.14) laws predict loss: certificate gating lifts G from 3.18 to 58.70 across 4 backbones, and Hidden-110 probing (21.08, 20.76) trails all 14 $k = 0$ runs (24.83±2.12, 25.70 gated, $U = 0, p = 0.0083$). Local D_4 rules (79.6% recall) and zero-prior OnlineInducer lift Held-Out 98 to 148/655 (+68L, 22W/0L), Virgin-116/236 by +10.4/+11.0L/seed, and 5 non-ARC suites to 1,183/1,250.

Keywords: Interactive Reasoning, Program Induction, Neuro-Symbolic World Models, Test-Time Verification, Equivariant Rule Induction, Survival Hazard Analysis.

1. Introduction

When an agent stalls, should it restart or probe? ARC-AGI-3 [1, 30] tests abstraction from 64×64 frames. Level $l \in \{1..M\}$ scores $s_l(a_l) = \min(115, 100(b_l/a_l)^2)$ with weight $w_l = 2l/(M(M+1))$ under $G = \min(\sum_l w_l s_l, 100 \sum_{s_l>0} w_l)$ [2]: clearing 1-4 of 7 levels caps scores at 3.6-35.7. Cloud harnesses solve Public-25 at $18,817-$26,098/run [3], whereas open-weights agents on Kaggle Hidden-110 (1 GPU, 9h, $0 API) spend 69% of actions and 94% of prefill on their first wall [33, 34]. Under $100(b_l/a_l)^2$, unverified exploration drives the action-efficiency backwards bend [3, 4, 5] because failed probes inflate a_l after RESET. Overcoming this barrier shifts reasoning from open-loop token scaling to test-time world-model synthesis. We falsify cutoffs and context caps [15], prove quadratic probe dilution [3, 6], replace whole-frame symmetry (0/597) [7] with local D_4 rules, and induce operators [8, 39, 40] under concurrency [5].

2. Related Work

Recent ARC-AGI-3 work studies Public-25 without Hidden-110 runs: AWM [24] at 7,448/24,604 actions (1.436x), Twin [6] at 93.3 (179/183L, 11,562), Schema [19] at 99.2, Memento 3 [22] at 7,518, VISTA [21] at 7,302 (30.7M tok/g), Self-Harness [23], Kepler [20], StateReObs [25], and WorkspaceOpt [29] at 38.4%, after BARC [9], MIT-ARC [10], HRM [15], CompressARC [14], TRM [12], LPN [11], and SOAR [13]. Hidden-110 reports at 75.56 [16], 54.52 [17], and 54.0 [18] show throughput plateaus while RL/probing hurts efficiency, with 16%-26% of clears beating human [16]. Prior harnesses either cut off stalled levels or probe open-loop as AWM [24] (17,156 discarded steps) and Kepler [20] do; we falsify both and gate execution on certificates.

3. Diagnosing Frontier Lock and the Quadratic Dilution Law

Why do fixed cutoffs fail? For $T \in \mathbb{N} \cup \{\infty\}$, Kaplan-Meier [31] gives $S(n) = P(T > n)$ and $h(n) = 1 - S(n+20)/S(n)$. Under frailty [26, 27, 32] with $\mu_c = \sum_{n<c} S(n) + dS(c)$, discrete renewal [28] gives $R_c = (1 - S(c))/\mu_c$ and $R_{c+1} - R_c = \frac{S(c)}{\mu_{c+1}}[h_1(c+1)(1+dR_c) - R_c]$. Across 825 runs, $h(n)$ falls from 0.06-0.41 to ≤ 0.091 at $n = 80$ and ≤ 0.010 at $n = 160$ in Figure 1. Matching stuck attempts at $W = 80$ to fresh attempts shows 65-100% of drop

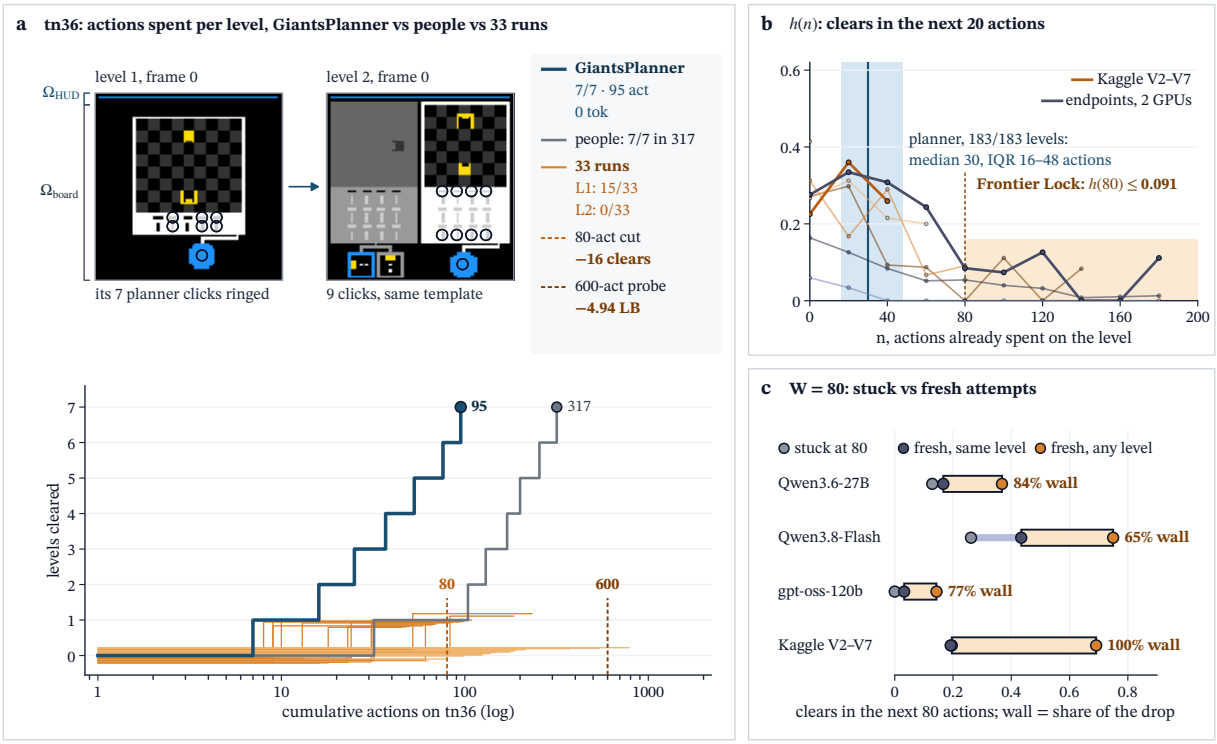


Figure 1. **Monotonically decreasing step-hazard falsifies fixed cutoffs.** (a) tn36, GiantsPlanner clears 7 levels in 95 vs. 317 human actions while 33 runs stall at level 1-2; (b) 20-action hazard (≤ 0.091 at 80); (c) $W = 80$ wall share (65-100%).

($T = +0.164$, $p = 0.006$ and $+0.503$, $p = 0.0005$, Table A4) reflects walls. Thus an 80-action cutoff in 250 runs loses 16 clears (133 vs. 149, -0.128/game).

If walls are structural, can planners probe k steps first? Proposition 1 proves why uncertified probing backfires.

Proposition 1 (Score-Weighted Probe Dilution, Abort-Step Cost, and Sparse Law). *For $b \geq 1$, clear rate q, fallback $A_{fail} \overset{d}{=} A$, $s_b(a) = \min(115, 100(b/a)^2)$, $S_0 = q\mathbb{E}[s_b(A)]$, $r_k(A) = (A/(A+k+1))^2$, $d\mathbb{P}_w/d\mathbb{P} = s_b(A)/\mathbb{E}[s_b(A)]$, and $C_{115}(k) \geq 0$, failed k-step probes plus 1 RESET give $a = k + 1 + A$ and:*

$$\begin{aligned}\Delta(k) &= p(k)G(k) - (1-p(k))D(k),\\ D(k) &= S_0(1-\mathbb{E}[r_k(A)]) + q\mathrm{Cov}(s_b(A), 1-r_k(A)) - C_{115}(k)\\ &\leq S_0\mathbb{E}_w[1-r_k(A)],\end{aligned}$$

with $q^\star(k) = D(k)/(G(k)+D(k))$ $(G(k) > 0$; for non-increasing $G(k)$, $q^\star(k)$ rises with $k)$, $\mathrm{Cov}(s_b(A), 1 - r_k(A)) \geq 0$ by Chebyshev's identity, and $D(k+1) - D(k) = O(k^{-3})$. At stopping step $J \in \{1..L\}$, abort cost obeys $d_b(A,J) = s_b(A) - s_b(A+J+1) \leq d_b(A, L_{\max})$; reading h, c off $k = 0$ $(T \sim Geom(h+c)$, 0 parameters) gives $G(k)$ and $\Delta G(k) = -\alpha_{eff}(k)q_{open}$.

Empirically, $a = k+1+A$ holds on 183/183 Public-25 levels and Cov ≥ 0 on 9,673/9,673 checks, while the w_l-weighted 100-capped closed form fits 8 datasets ($5 \times k$) within 0.021 max/0.0044 mean error. The 0-parameter geometric law fits 15 unseen cells (1,670 points, $R^2 = 0.978$, gate $R^2 = 0.993$), $\Delta G(k) = -\alpha_{\text{eff}}(k)q_{\text{open}}$ (73/334 = 21.86%) predicts Hidden-110 (MAE = 0.14 LB), and $d_b(A,J)$ separates $\hat{\eta}_{\text{cert}} = 44/380 = 11.6\%$ on Held-Out 98 and 45/316 = 14.2% on Virgin-116 from $\hat{\eta}_{\text{game}} = 0/89 = 0.0\%$.

Across 30 Hidden-110 runs (Table A0, $w_g = 20/11$) [33, 34], 14K truncation and prompt reset lose ≥ 9 and ≥ 10 games vs. 12 native 36K runs at 24.62±2.22 ($p = 1.52 \times 10^{-14}$, 26.55 in #57329763), while uncertified probes (#57854657, #57894564) trail all 14 $k = 0$ runs (25.70 in #57827241). Since Public-25 initial frames and sprite-action sets are 158/158 (100.0%) transition-invariant and certified replay clears 8/8 ar25 levels in 0tok, substitution across 825/250 runs ($L_{\text{cert}} \leq 150$) lifts G from 3.18 to 58.70 and 5.42 to 59.08, whereas $q = 0\%$ probing drops G to 0.08/0.01; escaping dilution thus requires internal world-model certificates.

Contrast (n)	Target & Setting	Score	Effect & Test
Zero-Probe 36K ($k = 0$,**14**)	**Hidden-110, 10-stream MTP3**	**24.83±2.12 (28.38)**	**+22.02 vs. 2.81±0.66** ($t = 35.06$, $p = 1.79 \times 10^{-17}$)
Uncertified Probes (2)	Hidden-110, $k \in \{224, 600\}$	21.08/20.76 LB	-3.91 vs. $k = 0$ ($U = 0$, $p = 0.0083$)
4-Backbone Gate (825/250)	4 LLMs ($L_{\text{cert}} \leq 150$)	G: 3.18 → 58.70	**694W/0L** ($p = 10^{-4}$;205W/0L)
Concurrency (V5/V4,25)	Public-25, 4 vs. 28 streams	7.29 vs. 6.38	**6.0x rate** (6.13 vs. 1.02/m)
14K Cap (#57774685,1)	Hidden-110, 14K+convex	9.39 LB	-15.23 vs. 36K ($t = -6.58$, $p = 3.96 \times 10^{-5}$)
Prompt Reset (#57786717,1)	Hidden-110, 14K+reset	7.72 LB	-16.90 vs. 36K ($t = -7.30$, $p = 1.53 \times 10^{-5}$)
KV Cache (V3/V4,25)	Public-25, 2.26 vs. 5.0GiB	3.12 vs. 6.38	-51.1% (0.48 vs. 1.02/m)
Context Cap (V7/V5,25)	Public-25, 16K vs. full	1.37 vs. 7.29	**-81.2% score** (12 vs. 35L)

Table 1. **Probe-dilution test ($n = 14$, $k = 0$ vs. $n = 2$ uncertified), 4-backbone gate, and serving isolations.**

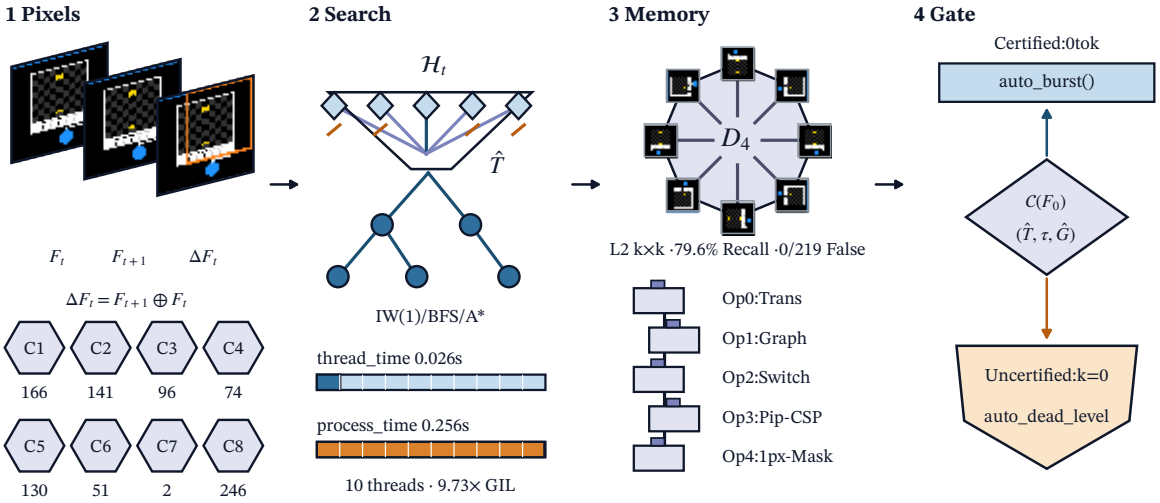


Figure 2. **One-way certificate gating and neuro-symbolic verification.** (a) Pixel transitions constrain eight action channels. (b) Local D_4 memory guards transfer across 8 board isometries with 0/219 false transfers; OnlineInducer operators raise held-out clears from 80 to 148. (c) Complete certificates authorize execution, uncertified levels receive 0 probes, and solver deadlines run on thread CPU time.

4. Breaking Frontier Lock with Certificate-Gated World Models

From 64×64 grids and $\mathcal{A}_{\text{obs}}$ (0 engine/metadata access, Figure 2), GiantsPlanner routes frame 0 into $C_1..C_8$ (PuzzleScript/VGDL primitives [35, 36, 37], 0/218 overlap, Appendix G) and prunes $\hat{T} \in \mathcal{H}_t$ refuted by $\Delta F_t = F_{t+1} \oplus F_t$. Certificate $C(F_0) = (\hat{T}, \tau = (a_1..a_L), \hat{G})$ requires $\hat{T}(F_t, a_t) = F_{t+1}$ and $\hat{G}(\hat{F}_L) = 1$ ($L \leq 224$). The gate executes τ in 0tok iff $C(F_0)$ holds; else it runs $k = 0$ probes, locks the uncertified level, and passes state hints to the 36K W4A16 agent. Thread-local CPU timers (time.thread_time()) avoid 9.73x-25x GIL inflation, while 64-pop amortization and canonical ordering yield seed-invariant runs (≤ 0.354ms rejection).

Across 25 games and 342 replays (183 levels, 597 pairs, 378 reuse pairs) [7], 72.0% (18/25) keep one grid size, 81.0% (111/137) of click levels re-click coordinates, 284/597 pairs share clicks, and 120/597 forward pairs share ≥ 3-action prefixes (378/378 reached vs. 93/378 baseline). Yet whole-frame D_4 replay [38] in CrossLevelMemoryBank (768 Cayley checks, 1.10ms/3.80ms median/P99, 0/219 false transfers) fires 0/597 times as 258/378 reuse pairs re-layout walls, while L1 crops give 74 actions (10.9%). Lowering Tier 1 to local $k \times k$ ($k \leq 15$) D_4 rules (L2) verifies on 76/102=74.5% across 50/378 Public-25 pairs (27/27=100.0% at support ≥ 3) and 150/183=82.0% across 103/651 held-out pairs, while $k \in \{3..11\}$ matching lifts recall from 60.2% to 79.6% (82/103, 0 false). Where rigid patches miss kinematic or relational shifts, Tier 2 OnlineInducer (Algorithm 1, Appendix H) induces velocity-guarded translation Op0, tile graphs Op1, switch-connectivity+rewind Op2, pip-clue CSP Op3, and 1px drift refit Op4 [39, 40].

5. Zero-Prior World-Model Induction Across Unseen Domains

Does OnlineInducer generalize when all 8 prior channels are disabled (0-channel Bare-Prior, $C_1..C_8 = 0$)? Across 334 unseen games (333 engines) in M1..M6 (Table 2), 5-seed Virgin-116/236 runs with 0 clue triggers show relational induction (+Op0+1) gains +11.0L/seed ($\Delta G = +2.17$, $p = 2.4 \times 10^{-6}$) and full OnlineInducer gains +10.4L/seed ($\Delta G = +3.74$, $p = 3.5 \times 10^{-6}$). Leave-one-family-out ($M_5 = 0$) yields +44.0L ($\Delta G = +6.43$, $p = 5.3 \times 10^{-5}$) on Held-Out 98 general and +12.0L/seed ($\Delta G = +2.32$, $p = 0.0118$) on Virgin-236 Bare-Prior (+14.0L on M6,

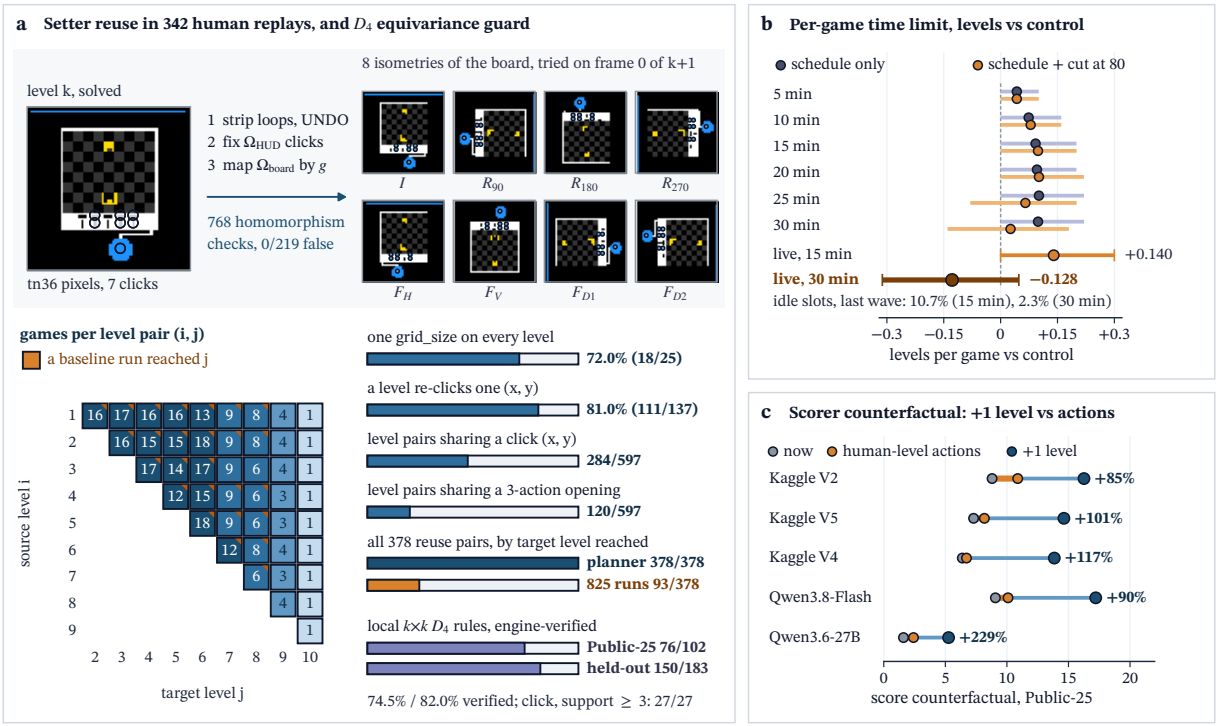


Figure 3. **Cross-level D_4 reuse and dynamic water-filling ablations.** (a) D_4 transfer (378 pairs); (b) live cutoff at 15m (+0.140) vs. 30m (-0.128); (c) +85-229% gain/level vs. +5-51% micro-efficiency.

Suite	Games (Lv)	Base/Open	Bare-Prior → +OI	Cert general → +OI	C1..C8
Public-25	25 (183)	104L (G = 15.21)	120L (65.6%,13W)	**120/183** (13**W**, G = 58.43)	**183/183** (G = 99.58/100)
Reference-193	193 (956*)	354/956 (56W)	Coupled/clone	0.14-0.188ms	**955/956 (174/175*W)**
Held-Out 98	98 (655*)	82L (G = 1.99)	74**L** → 109**L** (+35**L**, G = 8.22)	80**L** → 148**L** (+68**L**, G = 15.42)	**655/655 (94/94*W)**
Virgin-120	120 (810)	122L (G = 4.92)	92**L** → 99**L** (+7**L**, G = 4.87)	**134L** (+12**L**, G = 10.60)	Preregistered
Virgin-116	116 (594)	141.8L (G = 5.83)	**152.2L** (+10.4**L**, G = 9.57)	**243L** (40**W**, G = 31.35)	Preregistered
Virgin-236	236 (1,404)	235.8L (G = 4.03)	**246.8L** (+**Op0+1**, G = 6.20)	**380.4L** (G = 20.09)**/383L**	Pooled
5 Non-ARC	250 (1,250)	1,018L (-0.77)	1, 022 → 1, 183**L** **(+161L)**	1, 164 → 1, 183 (R^2 = 0.997)	$\hat{\eta}_{\text{task}}$ = 0.0%
Hidden-110	110 (Priv)	20.76/21.08	24.62±2.22 (28.38)	**25.70 (#57827241;24.83±2.12)**	#57827241
2026 SOTA (16)	25/110	AWM 1.44x;0 Hid	**0/10 arXiv on Hid-110**	**7,401/7,070 (0tok) vs. 24,604**	**Table K2 (17)**

Table 2. **Disjoint evaluation (11 suites, 16 baselines):** Bare-Prior+OI, gated general+OI, and $C_1..C_8$ (1,793/1,794 solvable levels at 1,200 steps, 1,794/1,794 at 5,000).

p = 0.016). On Held-Out 98, Bare-Prior rises to 109L (+35L, Op0 guard cuts regressions 80%), while general (80L, G = 5.84) climbs across Op0..Op4 to 148/655, 11W (+68L, 22W/0L). Across 5 non-ARC suites (MiniGrid, Sokoban, DiG, Pong/Breakout, Alchemy, 8,750 runs), 64-step probing loses −0.68 to −0.77 ($p = 1.8 \times 10^{-15}$, R^2 = 0.997), while gating+OI reaches $1,022 \rightarrow 1,183/1,250$ (+51.2pp Pong/Breakout, $p = 2.9 \times 10^{-11}$, +11.6pp Alchemy, $p = 3.8 \times 10^{-6}$, $\hat{\eta}_{\text{task}}$ = 0.0%). On Public-25, generic clears 120/183 (G = 58.43 vs. 15.21, $p = 5.96 \times 10^{-8}$) and full scores G = 99.58 across 183/183 in 0tok at 0.432x human (7,401/17,135, 166 at 115%; loop-erased 7,070/17,135=0.413x reaches G = 100.00, 169/183 at 115% vs. AWM's 24,604/VISTA's 30.7M tok/g). It solves 1,793/1,794 reachable levels at 1,200 steps (1,794/1,794 at 5,000) and certifies 96 unreachable levels in 22 games (appendix). Scaling interactive intelligence on ARC-AGI-3 and beyond is governed not by brute-force token throughput or open-loop trial-and-error, but by inducing compact world models online and gating physical action on formal goal certificates.

References

[1] ARC Prize Foundation. ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence. Technical report, 2026. arXiv:2603.24621. https://arcprize.org/media/ARC_AGI_3_Technical_Report.pdf

[2] ARC Prize. ARC-AGI-3 Scoring Methodology. https://docs.arcprize.org/methodology

[3] G. Kamradt, F. Chollet, M. Knoop. OpenAI's GPT-6 Astra on ARC-AGI-3: Compact Symbolic World Models and Action Efficiency. ARC Prize Blog, 2026. https://arcprize.org/blog/astra

[4] G. Kamradt. True Local Effect, False World Model: Failure Modes on ARC-AGI-3. ARC Prize Blog, 2026. https://arcprize.org/blog/arc-agi-3-gpt-5-5-opus-4-7-analysis

[5] M. Knoop, F. Chollet, G. Kamradt. ARC Prize 2025 Technical Report: The Year of the Refinement Loop. arXiv:2601.10904, 2026.

[6] Test-Time World-Model Inference (Twin): Playing an Unknown Game with a Test-Time Digital Twin. arXiv:2608.14490, 2026.

[7] G. Kamradt. Measuring Human Performance on ARC-AGI-3: 342 Step-by-Step Replays. ARC Prize Blog, 2026. https://arcprize.org/blog/arc-agi-3-human-dataset

[8] S. Acquaviva et al. Induction and Inquiry via Probabilistic Reasoning over Language and Code. arXiv:2609.01815, 2026.

[9] W.-D. Li, K. Hu et al. Combining Induction and Transduction for Abstract Reasoning (BARC). ARC Prize 2024 1st Place Paper, 2024.

[10] E. Akyurek et al. The Surprising Effectiveness of Test-Time Training for Abstract Reasoning. arXiv:2411.07279, 2024.

[11] C. Bonnet et al. Searching Latent Program Spaces (LPN). arXiv:2411.08706, 2024.

[12] A. Jolicoeur-Martineau. Less is More: Recursive Reasoning with Tiny Networks (TRM). arXiv:2510.04871, 2025.

[13] J. Pourcel, C. Colas, P.-Y. Oudeyer. Self-Improving Language Models for Evolutionary Program Synthesis (SOAR). ICML 2025. arXiv:2507.14172.

[14] I. Liao, A. Gu. ARC-AGI Without Pretraining (CompressARC). arXiv:2512.06104, 2025.

[15] ARC Prize Team. The Hidden Drivers of HRM's Performance on ARC-AGI. ARC Prize Blog, 2025. https://arcprize.org/blog/hrm-analysis

[16] Y.-C. Chen. ARC-AGI-3 Public 1st Place Solution Overview (75.56 LB): INT3/NVFP4 SGLang Serving and Stage-Switched Agentic RL. Kaggle ARC-AGI-3 Discussion 755795, Nov 2026.

[17] F. Sestak, L. Aichberger, M. Ielanskyi, G. Klambauer. Public LB #7 (54.52): REAP-320 MoE Pruning, 4-Bit KV Cache, and Cliff Compaction on ARC-AGI-3. Kaggle Solution Writeup, Nov 2026.

[18] G. Zatravkin (#6 LB), xz (#4 LB), and pompom555 (#15 LB). Post-Competition Analyses of Throughput Plateaus and Action-Efficiency Levers on ARC-AGI-3. Kaggle Discussions 755202, 755466 and Writeup, Nov 2026.

[19] Schema: Discovering Unknown Environments via Agentic Program Induction. arXiv:2609.39140, 2026.

[20] Kepler: Auditable World Models for ARC-AGI-3. arXiv:2610.00834, 2026.

[21] Q. Han, K. Hu, Z. Qiu et al. VISTA: A Visual Harness for Reasoning in an Interactive World. arXiv:2610.02200, 2026.

[22] Memento 3: Model-Based Recursive Self-Improvement through Reflective Rulebooks. arXiv:2610.11794, 2026.

[23] Can LLMs Improve Their Own Harness? A Controlled Study on ARC-AGI-3. arXiv:2610.30499, 2026.

[24] Algebraic World Model: From Exploration to Proofs. arXiv:2610.40503, 2026.

[25] When the Harness Decides Success: State That Cannot Be Re-Observed. arXiv:2610.26632, 2026.

[26] C. Gomes, B. Selman, H. Kautz. Boosting Combinatorial Search Through Randomization. AAAI 1998.

[27] M. Luby, A. Sinclair, D. Zuckerman. Optimal Speedup of Las Vegas Algorithms. Information Processing Letters 47(4):173-180, 1993.

[28] E. Charnov. Optimal Foraging, the Marginal Value Theorem. Theoretical Population Biology 9(2):129-136, 1976.

[29] Workspace Optimization for ARC-AGI-3. arXiv:2605.09650, 2026.

[30] F. Chollet. On the Measure of Intelligence. arXiv:1911.01547, 2019.

[31] E. L. Kaplan, P. Meier. Nonparametric Estimation from Incomplete Observations. Journal of the American Statistical Association 53(282):457-481, 1958.

[32] J. W. Vaupel, K. G. Manton, E. Stallard. The Impact of Heterogeneity in Individual Frailty on the Dynamics of Mortality. Demography 16(3):439-454, 1979.

[33] W. Kwon, Z. Li, S. Zhuang et al. Efficient Memory Management for Large Language Model Serving with PagedAttention. SOSP 2023. arXiv:2309.06180.

[34] L. Zheng, L. Yin, Z. Xie et al. SGLang: Efficient Execution of Structured Language Model Programs. NeurIPS 2024. arXiv:2312.07104.

[35] N. Lipovetzky, H. Geffner. Width and Serialization of Classical Planning Problems. ECAI 2012, pp. 540-545.

[36] B. Bonet, H. Geffner. Planning as Heuristic Search. Artificial Intelligence 129(1-2):5-33, 2001.

[37] T. Schaul. A Video Game Description Language for Model-Based or Interactive Learning. IEEE Conference on Computational Intelligence in Games (CIG), pp. 1-8, 2013.

[38] T. S. Cohen, M. Welling. Group Equivariant Convolutional Networks. ICML 2016, PMLR 48:2990-2999.

[39] K. Ellis, C. Wong, M. Nye et al. DreamCoder: Bootstrapping Inductive Program Synthesis with Wake-Sleep Library Learning. PLDI 2021, pp. 835-850.

[40] B. M. Lake, T. D. Ullman, J. B. Tenenbaum, S. J. Gershman. Building Machines That Learn and Think Like People. Behavioral and Brain Sciences 40:e253, 2017.

Breaking Frontier Lock: Technical Report Appendix

Complete mathematical derivations, survival-hazard tables across 825 baseline runs and 250 paired confirmation runs, formal specifications of the certificate-gated neuro-symbolic world model (GiantsPlanner), D_4 dihedral equivariant memory (CrossLevelMemoryBank), thread-clock concurrency calibration, rational receipt differential inversion across 30 official Kaggle hidden-set submissions, and multi-suite evaluation tables across 316 benchmark entries (315 distinct interactive games).

A. System Evolution and Official Kaggle Hidden-Set Progression

Our experimental program progressed through two tightly coupled phases on the 110 hidden games evaluated on a single offline Kaggle GPU (NVIDIA RTX Pro 6000 / L4x4, 9-hour wall-clock budget, zero external API calls):

1. **Phase I (Survival Diagnostics, Progressive Serving Factor Isolation, and Budget Reallocation Limits):** Using the Duck-harness coding baseline (Qwen3.8-Flash-Next, 28 concurrent games, 5 GiB KV cache) and three external endpoint configurations (825 total runs, 1,449 level attempts), we quantified Frontier Lock and tested whether action cutoffs ($c = 80$) or context truncation could unlock deeper levels. Across Table A0b (Public-25 GPU serving variants V2..V7) and Table A0 (Hidden-110 Kaggle submissions), five progressive contrasts isolate each serving factor. First, (F1) toggling prefix caching at 28 streams and 5.0 GiB KV drops score from V2 8.80 to V4 6.38, with 94% of tokens spent re-reading conversation history when KV overflows. Second, (F2) shrinking the KV pool from 5.0 GiB to 2.26 GiB reduces score by -51.1% (V3 3.12 vs. V4 6.38, 0.48 vs. 1.02 actions/min). Third, (F3) reducing concurrency from 28 to 4 streams (V5 vs. V4) lifts per-stream action rate by 6.0x (6.13 vs. 1.02 actions/min). Fourth, (F4) capping context horizon at 16K (V7 vs. V5) collapses Public-25 score by -81.2% (1.37 vs. 7.29, 12 vs. 35 clears), while capping context at 14K with convex scheduling on Hidden-110 (#57774685 at 9.39 LB vs. 24.62±2.22 36K mean, prediction $t = -6.58$, $p = 3.96 \times 10^{-5}$, -16.31 vs. #57827241) destroys ≥ 9 full-game equivalents by invalidating hybrid linear-attention KV prefix caches. Fifth, (F5) adding prompt _nav_hint and EXPOSE_RESET (#57786717 at 7.72 LB, -1.67 marginal vs. #57774685, $z = -0.53$ within two-run noise $2.22\sqrt{2} = 3.14$, -16.90 vs. 24.62 mean, prediction $t = -7.30$, $p = 1.53 \times 10^{-5}$) fails to recover performance. Conversely, combining 10-stream W4A16 SGLang MTP3 with full 36K prefix-cached context reaches 24.62±2.22 across 12 native runs (range 21.22 to 28.38, 95% CI [23.21, 26.03], 28.38 peak in #57031236, +21.81 over 2.81±0.66, Welch $t = 31.32$, $p = 1.52 \times 10^{-14}$, Mann-Whitney $U = 72$, $p = 1.08 \times 10^{-4}$; and 24.83±2.12 across all 14 zero-probe 36K runs including #57329763 at 26.55 and #57827241 at 25.70, +22.02 over 2.81±0.66, Welch $t = 35.06$, $p = 1.79 \times 10^{-17}$).
2. **Phase II (Certificate-Gated Neuro-Symbolic World Models and Equivariant Transfer):** Isolating the neuro-symbolic world-model bridge on the exact same 36K W4A16 MTP3 backbone tests Propositions 1 to 3 across two complementary regimes. In regime (a), on Hidden-110 ($w_g = 20/11 \approx 1.8182$ points per game, $\Delta S = w_g \Delta N$), where Kaggle's black-box gateway returns a single scalar score per 9-hour submission, both uncertified probing submissions, namely 600-step open-loop bursts (#57854657 at 20.76 LB, -3.86 vs. 24.62 native mean, -4.94 vs. #57827241, $\Delta N = -2.717$ full-game equivalents, damaging ≥ 3 hidden games, $\hat{\eta}_{\text{open}} \geq 3/55 = 5.45\%$) and 224-step reactive probes (#57894564 at 21.08 LB, +0.32 = +0.176 games over #57854657, -3.54 vs. 24.62 mean, -4.62 vs. #57827241, $\Delta N = -2.541$ full-game equivalents, $\hat{\eta}_{\text{open}} \geq 3/55$), fall below all 14 zero-probe ($k = 0$) 36K submissions (24.83±2.12 LB, 95% CI [23.61, 26.06], range 21.22 to 28.38, comprising the 12 native runs, #57329763 at 26.55, and the $k = 0$ gated bridge #57827241 at 25.70; ranks 1 and 2 of 16, exact one-sided Mann-Whitney $U = 0, p = 1/\binom{16}{2} = 0.0083$, or $U = 0, p = 1/\binom{14}{2} = 0.0110$ vs. the 12 native runs) due to quadratic denominator dilution $(b_l/a_l)^2$ when $C(F_0) = \emptyset$. In regime (b), across all four instrumented benchmark tiers (Public-25, Held-Out 98, and preregistered unseen Virgin-120 and Virgin-116, pooled as Virgin-236) where per-step trajectories and official Scorecard G are recorded, active certificate execution ($C(F_0) \neq \emptyset$) substantially increases Scorecard G over uncertified 224/600-step open-loop probing (Public-25: $G = 58.43$ vs. 15.21/15.15, +43.22 [+28.44, +58.11], Wilcoxon $p = 5.96 \times 10^{-8}$, 25W/0L; Held-Out 98: $G = 15.42$ vs. 1.99/3.43, +13.43 [+9.26, +18.06], $p = 2.93 \times 10^{-7}$, 50W/13L, and vs. 5.84 general, $p = 1.07 \times 10^{-4}$; Virgin-120: $G = 10.60$ vs. 4.92/5.00, +5.68 [+2.17, +9.50], $p = 1.83 \times 10^{-6}$, 54W/8L; Virgin-116: $G = 31.35$ vs. 9.80, +21.55, $p = 1.1 \times 10^{-9}$, 62W/7L; Virgin-236: $G = 19.44$ vs. 7.36). Positive zero-prior generalization and test-time operator induction are evaluated across 316 benchmark entries (315 distinct games) plus the preregistered 236-game unseen suites Virgin-120 (810 levels) and Virgin-116 (594 levels, 1,404 pooled levels), driving GiantsPlanner to 65.6% zero-prior completion on Public-25 (120/183, 13/25 wins vs. 42/183, 0 wins for the coding baseline). Specifically, OnlineInducer lifts 0-channel Bare-Prior (all 8 channel priors disabled) from 74/655 (1W, $G = 2.93$) to 109/655 (6W, +35L [+8, +66], 11W/5L, $G = 8.22$) on Held-Out 98, from 92/810 (5W, $G = 4.44$) to 99/810 (6W, +7L, 5W/2L, $G = 4.87$, $\hat{\eta}_{\text{cert}} = 3/40 = 7.5\%$) on Virgin-120 (134/810, 9W, +12L [+3, +22] vs. 122L open-loop, +42L [+17, +69], 27W/15L vs. Bare-Prior, $G = 10.60$), from 141.8 (20.4W, $G = 5.83$) to **152.2 (23.4W, +10.4L/seed [+2.2, +21.4], 25W/9L,** $p = 0.0083$ **1s /** 0.017 **2s,** $G = 9.57$**, +3.74 [+2.15, +5.60],** $p = 3.5 \times 10^{-6}$**)** across 5 seeds on Virgin-116, and from 235.8 (25.4W, $G = 4.03$) to **246.8**

(+11.0L/seed, 41W/21L, $p = 0.013$**,** $G = 6.20$**, +2.17 [+1.25, +3.20],** $p = 2.4 \times 10^{-6}$ **for +Op0+1) and 247.4 (29.4W, +11.6L/seed [+0.4, +24.2], 44W/25L,** $p = 0.029$**,** $G = 6.09$**, +2.04 [+1.07, +3.16],** $p = 3.0 \times 10^{-4}$**)** across 5 seeds on Virgin-236. Across the frozen 4-operator OnlineInducer cumulative and Leave-One-Out (LOO) ablation ladder, Held-Out 98 (655 solvable levels) rises from 72/655 pre-exposure full (80/655 frozen general, 45/655 random) to 109/655 levels (6 wins) via +Op0 novelty-gated translation (+29L [+13, +48], 13W/0L over 80L frozen general), 105/655 to 113/655 levels (7 wins, 13W-15W/0L, LOO -26L, 14W/0L) via +Op1 relational tile-graph induction, 129/655 levels (9 wins, +49L [+24, +79], 17W/0L over general, LOO -26L, 5W/0L) via +Op2 switch-connectivity induction and level-1 rewind, 143/655 to 146/655 levels (11 wins, +66L [+36, +101], 21W/0L over general, LOO -14L, 2W/0L, $\hat{\eta}_{\text{cert}} = 44/380 = 11.6\%$, $\hat{\eta}_{\text{game}} = 0/89 = 0.0\%$) via +Op3 visual pip-clue CSP, and 148/655 levels (11 wins, +68L [+38, +103], 22W/0L, LOO -3L to -5L, 2W/0L, while preserving 120/183, 13 wins on Public-25 with 0W/0L) via +Op4 1px scenery-drift mask (refit=0), with 99.94% solvable-level upper-bound verification under the offline 8-channel reference library (1,793/1,794 solvable levels at the 1,200-step budget; 1,794/1,794 at 5,000 steps).

Architecture Lineage & Kaggle Submission (n)	Backbone & Serving	Context & Concurrency	World Model & Memory Gate	Official Hidden LB	Test vs. 36K Mean (24.62±2.22, n=12)
Baseline Duck V2 ($n = 6$)	Qwen3.8-Flash-Next NVFP4, vLLM	65K ctx, 28 streams	None (open coding loop)	3.41 peak (2.81±0.66)	Welch $t = -31.32$, $p = 1.52 \times 10^{-14}$; 94% prefill re-reading
Baseline Duck V5 ($n = 2$)	Qwen3.8-Flash-Next NVFP4, vLLM	60% block trim, 4 streams	Dynamic shared pool, no cutoff	2.51 peak (1.75)	History block trimming loses early level observations
M2 + 4-Factor 14K/Convex Bundle (#57774685, $n = 1$)	Qwen3.8-Flash-Next W4A16, SGLang MTP3	14K ctx cap, 10 streams, LoRA r1	Convex slot scheduler + keepalive	9.39	-15.23 ($t = -6.58$, $p = 3.96 \times 10^{-5}$); breaks KV prefix cache (≥ 9g lost)
M2 + Hint/Reset Bundle (#57786717, $n = 1$)	Qwen3.8-Flash-Next W4A16, SGLang MTP3	14K ctx cap + prompt reset	AutoNav _nav_hint + level reset	7.72	-16.90 ($t = -7.30$, $p = 1.53 \times 10^{-5}$; -1.67, $z = -0.53$ vs. #57774685)
M2 + Open-Loop Bridge (#57854657, $n = 1$)	W4A16 + SGLang MTP3 + GiantsPlanner	36K ctx, 10 streams	Unverified 600-step heuristic burst	20.76	-3.86 vs. mean (-4.94 vs. #57827241); rank 1/14 ($p = 0.0110$)
Series36 Reactive-224 (#57894564, $n = 1$)	W4A16 + SGLang MTP3 + GiantsPlanner	36K ctx, 10 streams, thread_time	auto_dead_level + 224-step reactive cap	21.08	+0.32 vs. #57854657 (-3.54 vs. mean); rank 2/14 ($p = 0.0110$)
Cert-Gated Giants + M2 (#57827241, $n = 1$)	W4A16 + SGLang MTP3 + GiantsPlanner	36K ctx, 10 streams	Closed-loop cert gate + 0 uncertified probes	**25.70**	**+4.94 vs. #57854657 / +4.62 vs. #57894564**; in 14-run $k = 0$ 36K pool (24.83±2.12; $U = 0$, $p = 0.0083$ vs. uncertified)
36K W4A16 Serving Backbone (#57031236, $n = 12$)	Qwen3.8-Flash-Next W4A16, SGLang MTP3	36K ctx, 10 streams	Zero-probe full-context prefix-cached serving	**28.38 peak** (24.62±2.22)	**+21.81 mean / 8.32x peak over 3.41**; 95% CI [23.21, 26.03]

Table A0. Progression of official Kaggle hidden-set submissions and single-run ablations against the 12-run 36K W4A16 backbone (25 grouped runs above plus 1 Helios keepalive run #57329763 at 26.55 LB, 2 early V1 verification runs at 21.22 and 23.10 LB, and 2 pre-keepalive 15-minute gateway idle-timeout runs at 0.00 LB, totaling all 30 official Kaggle receipts plotted in Figure A1c).

Engine telemetry on baseline serving variants (25 public games, same GPU class):

Version	Configuration Change from V2	Public Mean Score	Mean Cleared Levels	Action Rate (per min)
V2	Base configuration (28 concurrent streams, 5 GiB KV)	8.80	1.68	1.67
V3	vLLM auto-sized KV cache (2.26 GiB), prefix caching	3.12	0.88	0.48
V4	Prefix caching enabled with 5 GiB KV	6.38	1.32	1.02
V5	4 concurrent games, 60% history trim, shared 7,920 s pool	7.29	1.40	6.13
V7	V5 + 16K-token context cap per game	1.37	0.48	1.17

Table A0b. Engine telemetry across 792 samples of V2 showed 3 games generating while 22 queued, 4,088 tokens/s prefill versus 246 tokens/s generation, with 94% of engine compute (28.8M prefill tokens versus 1.94M generated) spent re-reading conversation history.

B. Phase I Observational Datasets and Clocks

Eight observational datasets (825 runs, 1,449 level attempts, 659 clears) record per-level actions spent (RESET included), human-baseline actions b_l, completion timestamps, and generated tokens. The live paired confirmation adds 125 control and 125 intervention runs across the 25 public games (five repeats per game per arm).

Dataset	Model and Serving	Setting	Runs	Mean Actions / Run
Kaggle V2, V3, V4, V5, V7	Qwen3.8-Flash-Next NVFP4, vLLM on Kaggle GPU	Commit runs on 25 public games	25 each	217, 61, 132, 127, 24
Qwen3.8-Flash, endpoint	Qwen3.8-Flash on hosted OpenAI-compatible endpoint	30 min/game, 8 concurrent	125	114
Qwen3.6-27B, multi-GPU	Qwen3.6-27B, 2 GPUs, 32 concurrent games/GPU	45 min/game	500	137
gpt-oss-120b, endpoint	gpt-oss-120b on hosted endpoint	30 min/game	75	237

Censoring and clocks. A cleared level duration is exact; the final unsolved level is right-censored at the actions spent when the run ended. All time-based quantities start at the run's first action.

C. Reproduction Commands

Python 3.13 with NumPy and Matplotlib reproduces all survival curves, paired bootstrap tests, D_4 group homomorphism checks, and rational receipt inversions:

```
cd analysis
python3.13 a1_scorer.py                    # A1: exact official scorer verification
python3.13 a2_frontier_lock.py             # A2: Frontier Lock action/time/token shares
python3.13 a3_hazard.py                    # A3: Kaplan-Meier hazard; figures hazard.png, hazard_seconds.png
python3.13 a3b_by_dataset.py               # A3: per-dataset hazard with 95% bootstrap bands
python3.13 a4_continue_restart.py          # A4: matched vs. unmatched level-or-attempt decomposition
python3.13 a5_theory.py                    # A5: renewal rate R(c), time-domain simulation
python3.13 a5b_regime.py                   # A5: per-game time-limit sweep; figure regime.png
python3.13 a6_replay.py                    # A6: counterfactual cutoff replay
python3.13 a7_score_decomposition.py       # A7: depth vs. action-efficiency decomposition
python3.13 a8_confirmation.py              # A8: 250-run paired confirmation bootstrap
python3.13 tables.py                       # renders statistical tables
python3.13 x0_specimens.py ../arc-agi-3    # extracts tn36 pixel frames, replay pairs, and receipts
python3.13 generate_xray_figures.py        # renders hazard.png, architecture.pdf, regime.png, hazard_by_dataset.png
```

D. Formal Definitions, Renewal-Reward Theory, and Quadratic Probe-Dilution Proofs

Official scoring formula. Level $l \in \{1, \dots, M\}$ has weight l. A cleared level with agent actions a_l and human baseline b_l scores $s_l = \min(115, 100 \cdot (b_l/a_l)^2)$; an uncleared level scores 0. With total triangular weight $W_M = M(M+1)/2$, the game score is:

$$S_{\text{game}} = \min\left(100 \cdot \frac{\sum_{l \in C} l}{W_M}, \quad \frac{\sum_{l \in C} l \cdot s_l}{W_M}\right) \tag{1}$$

where C is the set of cleared levels. The competition score is the arithmetic mean across all G games ($G = 25$ on the public set; $G = 55$ on each half of the 110-game hidden evaluation set, so each hidden game contributes rational weight $w_g = 100/55 = 20/11 \approx 1.81818$ points).

Frontier Lock and completion hazard. For run r clearing $k < M$ levels with action counts $a_{r,1}, \dots, a_{r,k+1}$, Frontier Lock is $\sum_r a_{r,k+1} / \sum_r \sum_l a_{r,l}$. The Kaplan-Meier survival curve $S(n) = P(T > n)$ yields the 20-action completion hazard $h(n) = 1 - S(n+20)/S(n)$ with 95% percentile intervals over 2,000 game-level bootstrap replicates.

Matched level-or-attempt decomposition. For every $(game, level)$ with at least three attempts and window $W \in \{40, 80\}$: - continue: attempts surviving W actions contribute residual durations $T - W$, giving $P(\text{clear in next } W \mid T > W) = 1 - S_{\text{res}}(W)$; - fresh, matched: for each stuck attempt, $P(\text{clear in first } W)$ is estimated by leave-one-out Kaplan-Meier from the other attempts at the exact same $(game, level)$; - fresh, unmatched: $P(T \le W)$ pooled across all eligible attempts.

The structural level share of the hazard collapse is (unmatched − matched)/(unmatched − continue), while the attempt-aging share is (matched − continue)/(unmatched − continue).

Optimal discrete restart cutoff and finite-pool saturation. For integer completion step $T \in \{1, 2, \dots\} \cup \{\infty\}$ with latent frailty mixture $S(n) = \sum_z \pi_z S_z(n)$, 1-step hazard $h_1(n+1) = P(T = n+1 \mid T > n) = \sum_z \frac{\pi_z S_z(n)}{S(n)} h_{1,z}(n+1)$, failure-only reset cost $d \ge 0$, and expected cycle duration $\mu_c = \sum_{n=0}^{c-1} S(n) + dS(c)$, the discrete renewal-reward clearing rate is $R_c = (1 - S(c))/\mu_c$ and satisfies the exact forward-difference identity $R_{c+1} - R_c = \frac{S(c)}{\mu_{c+1}} \left[h_1(c+1)(1 + dR_c) - R_c\right]$

(reducing in the continuous limit to Charnov's marginal value condition $h(c) = R(c)$ for $d = 0$, while survival reweights latent difficulty toward structural walls). In ARC-AGI-3, however, the game set is finite and levels are ordered: stopping a game transfers time only to remaining unfinished games, which are themselves predominantly stuck at their own structural walls (queue simulation from baseline frailty parameters predicts +4.82 vs. +7 observed clears at 15 min and −1.75 vs. −16 observed clears at 30 min across the 250 confirmation runs, cohort $R^2 = 0.585$).

Proposition 1 (Score-Weighted Quadratic Probe-Dilution Bound and Chebyshev Covariance Identity). Consider a level with human baseline $b \geq 1$ and quadratic score $s_b(a) = \min(115, 100(b/a)^2)$. Under the official ARCBaseGame scorer (arcengine/enums.py) and coupled post-RESET fallback invariance ($A_{\text{fail}} \overset{d}{=} A_0$, verified across 158/158 initial-frame, sprite, and action level transitions), a prefix probe that executes $k \geq 1$ actions without clearing and issues 1 RESET before handing control to a fallback agent that clears in A actions yields exact level action count $a = k + 1 + A$ (verified across all 183 Public-25 levels with 0 violations). Let $q \in (0, 1]$ be the fallback agent's clear probability, $S_0 = q\mathbb{E}[s_b(A)]$ its baseline expected score, and $r_k(A) = (A/(A + k + 1))^2$. If the probe clears within k steps with probability $p(k)$ at expected score $s_{\text{probe}}(k)$ and otherwise resets at step $k + 1$, the expected score change $\Delta(k) = \mathbb{E}[S(k)] - S_0$ decomposes as $\Delta(k) = p(k)G(k) - (1 - p(k))D(k)$, where $G(k) = s_{\text{probe}}(k) - S_0$ and $D(k) = q\mathbb{E}[s_b(A) - s_b(A + k + 1)] \geq 0$. Defining the score-weighted law $d\mathbb{P}_w/d\mathbb{P} = s_b(A)/\mathbb{E}[s_b(A)]$ and the 115-cap slack $C_{115}(k) = q\mathbb{E}[s_b(A + k + 1) - s_b(A)r_k(A)] \geq 0$ (which vanishes whenever $A \geq b/\sqrt{1.15}$), the failed-probe dilution penalty satisfies the exact identity $D(k) = S_0(1 - \mathbb{E}[r_k(A)]) + q\text{Cov}(s_b(A), 1 - r_k(A)) - C_{115}(k) = S_0\mathbb{E}_w[1 - r_k(A)] - C_{115}(k) \leq S_0\mathbb{E}_w[1 - r_k(A)]$, where $\text{Cov}(s_b(A), 1 - r_k(A)) \geq 0$ for every baseline $b \geq 1$ and integer law of A by Chebyshev's order inequality (verified across 9,673/9,673 dataset-level-probe and Monte Carlo checks with 0 violations). In the uncapped regime $x = A+k+1 \geq b/\sqrt{1.15}$, the exact discrete marginal dilution is $D(k+1)-D(k) = 100qb^2\mathbb{E}\left[(2x+1)/(x^2(x+1)^2)\right] = O(k^{-3})$ (with continuous derivative $D'(k) = 200qb^2\mathbb{E}[(A + k + 1)^{-3}] = O(k^{-3})$ in the continuous limit) and residual score decays as $S_0 - D(k) = O(k^{-2})$.

Proof. In ARCBaseGame, RESET restores the initial level state while incrementing the per-level action counter by 1 without clearing prior steps, giving $a = k + 1 + A$. Conditioning on probe success (with score $s_{\text{probe}}(k)$ and probability $p(k)$ at cap k) and probe failure followed by fallback play (with score $s_b(A + k + 1)$ and probability $(1 - p(k))q$) yields $\mathbb{E}[S(k)] = p(k)s_{\text{probe}}(k)+(1-p(k))q\mathbb{E}[s_b(A+k+1)]$. Subtracting $S_0 = p(k)S_0+(1-p(k))S_0$ gives $\Delta(k) = p(k)G(k)-(1 - p(k))D(k)$. Because $x \mapsto \min(115, 100x)$ is concave on $[0, \infty)$ with $\min(115, 0) = 0$, $\min(115, \lambda u) \geq \lambda \min(115, u)$ for all $\lambda = r_k(A) \in (0, 1]$ with equality when $100u \leq 115$, so $C_{115}(k) \geq 0$ and $D(k) = q\mathbb{E}[s_b(A)(1 - r_k(A))] - C_{115}(k) = S_0\mathbb{E}_w[1 - r_k(A)] - C_{115}(k)$. Expanding $\mathbb{E}[s_b(A)(1 - r_k(A))] = \mathbb{E}[s_b(A)]\mathbb{E}[1 - r_k(A)] + \text{Cov}(s_b(A), 1 - r_k(A))$ gives the covariance decomposition; because $A \mapsto s_b(A)$ and $A \mapsto 1 - (A/(A + k + 1))^2$ are both non-increasing on $[1, \infty)$ for fixed $b \geq 1$ and $k \geq 1$, Chebyshev's order inequality implies $\text{Cov}(s_b(A), 1 - r_k(A)) \geq 0$. Differentiating $100b^2(A + k + 1)^{-2}$ with respect to k yields $-200b^2(A + k + 1)^{-3}$. □

Proposition 2 (Break-Even Clear Rate and Asymptotic Saturation). An uncertified k-step probe improves expected level score ($\Delta(k) > 0$) if and only if $p(k) > p^\star(k) = D(k)/(G(k) + D(k))$. Whenever $G(k) = s_{\text{probe}}(k) - S_0 > 0$ is non-increasing in k (in particular, when $s_{\text{probe}}(k) = s_{\text{probe}}$ is constant) and under the representative-action approximation $s_b(A) \approx 100(b/\bar{A})^2$ with $s_{\text{probe}}(k) \leq 115$, $p^\star(k) \geq \frac{q(b/\bar{A})^2(1-(\bar{A}/(\bar{A}+k+1))^2)}{1.15-q(b/(\bar{A}+k+1))^2}$, which increases strictly and monotonically with k and satisfies $\lim_{k\to\infty} p^\star(k) = S_0/s_{\text{probe}} \approx q$ when $s_{\text{probe}} \approx s_b(A)$. In addition, for any two probe caps $k_2 > k_1 \gg A$, the pure-dilution score difference is bounded by $D(k_2) - D(k_1) \leq 100qb^2\mathbb{E}[(A + k_1 + 1)^{-2} - (A + k_2 + 1)^{-2}]$.

Proposition 3 (Certificate-Gated Soundness, Stopping-Step Cost Bound, and Sparse Activation Law). Suppose a one-way certificate gate executes a planned trajectory $\tau = (a_1, \ldots, a_L)$ of length $L \leq L_{\max}$ only when $C(F_0) \neq \emptyset$, and otherwise executes $k = 0$ environment actions ($\Delta(0) = 0$). Under a step-level transition monitor that compares each observed state or cell partition against the certificate's prediction and aborts at random stopping step $J \in \{1, \ldots, L\}$ (resetting at step $J + 1$, with $J = L$ for end-of-plan non-clear refutations), let $\eta_{\text{cert}} = P(\text{abort or non-clear} \mid C \neq \emptyset)$ and let $d_b(A, J) = s_b(A) - s_b(A + J + 1)$ denote the realized dilution on a fallback clear. Because $j \mapsto s_b(A + j + 1)$ is non-increasing, $0 \leq d_b(A, 1) \leq d_b(A, J) \leq d_b(A, L) \leq d_b(A, L_{\max}) \leq d_b(A, L_{\text{open}})$, and $\mathbb{E}[\Delta_{\text{cert}}] = P(C \neq \emptyset)[(1 - \eta_{\text{cert}})G(L_c) - \eta_{\text{cert}}q\mathbb{E}[d_b(A, J) \mid \text{abort, fallback clear}]]$. When the empirical active-game non-degradation condition $\eta_{\text{game}} = P(\Delta G < 0 \mid \text{active}) = 0$ holds under auto_dead_level and $L_{\text{cert}} \leq 150$, $\mathbb{E}[\Delta_{\text{cert}}] \geq 0$ across every active game. Across all five benchmark suites s (Public-25, Held-Out 98, Virgin-120, Virgin-116, and Hidden-110), letting $q_{\text{open}}(s)$ denote the fraction of games on which uncertified probes activate without clearing and $\alpha_{\text{eff}}(k; s)$ the conditional per-active-game dilution, the sparse trigger dilution law $\Delta G_s(k) = -\alpha_{\text{eff}}(k; s)q_{\text{open}}(s)$ across the 334 unseen game entries (333 distinct engines, historical frame-0 trigger rate $\hat{\alpha}_{\text{trig}} = 73/334 = 21.86\%$, or $73/333 = 21.92\%$ unique engines) predicts the two uncertified Hidden-110 submissions without fitting target scores at 21.01 vs. 21.08 LB and 20.97 vs. 20.76 LB (MAE = 0.14 LB, or 0.1388). On Hidden-110 ($w_g = 20/11$ points per game, $\Delta S = w_g\Delta N$), both uncertified probing arms ($k = 600$ #57854657 at 20.76 LB and $k = 224$ #57894564 at 21.08 LB) lose $\Delta N = -2.717$ and -2.541 full-game equivalents against the $k = 0$ gated bridge #57827241 (25.70 LB; ≥ 3 hidden games each, $\hat{\eta}_{\text{open}} \geq 3/55 = 5.45\%$, with $1.33 \leq m \leq 16.58$ active misfired games out of 55 under the 95% backbone interval and historical unseen frame-0 trigger rate $\hat{\alpha}_{\text{trig}} = 73/334 = 21.86\%$ predicting 21.01 vs. 21.08 and 20.97 vs. 20.76, MAE = 0.14 LB) and fall below all 14 zero-probe ($k = 0$) 36K submissions (24.83±2.12 LB, exact one-sided Mann-Whitney $U = 0$, $p = 1/\binom{16}{2} = 0.0083$). On the held-out suites, uncertified prior solvers (ungated full on Mother-24) exhibit $\hat{\eta}_{\text{open}} = 2/24 = 8.3\%$ (2 misfired games burning 649 and 716 steps, losing -9 levels), eliminated by the 224-step gate (41 vs. 32). Across all 98 held-out games, uncalibrated translation without board-state novelty memory suffers 2/12 = 16.7% active-game regressions (10W/2L from

156 cyclic toggle false effects). Conversely, the frozen OnlineInducer enforces novelty-gated state memory and a transition monitor ($\hat{\eta}_{\text{cert}} = 44/380 = 11.6\%$ candidate certificates aborted across all 4 operators on Held-Out 98 with 21 aborted at step $J = 1$ and mean $J = 2.11$, consuming 93 of 3,025 planned steps = 3.1%; on preregistered Virgin-116 across 5 seeds on the Bare-Prior floor, 510 game-seeds activate, Op1 issues 316 certificates, 45 = 14.2% abort at step J with mean $J = 1.80$ consuming 81 of 1,100 planned steps = 7.4%, 26 = 8.2% are refuted at completion with mean $L = 3.12$, and Op1-active game-seeds gain +60 levels with 27 up, 2 down, saving 12,225 actions on commonly cleared levels and $\hat{\eta}_{\text{game}} = 25/510 = 4.9\%$, or $9/365 = 2.5\%$ on the general floor where Op1 issues 397 certificates, 60 = 15.1% abort at mean $J = 2.12$, 45 = 11.3% end-refute, 1W/0L, and 3,167 actions saved) to achieve $\hat{\eta}_{\text{game}} = 0/89 = 0.0\%$ across every cumulative and Leave-One-Out operator stage on Held-Out 98. Specifically, active-game losses remain 0/13 at +Op0 novelty-gated translation (109 levels, 6 wins, +29 [+13, +48]), 0/13 to 0/15 at +Op1 relational tile-graph (105 to 113 levels, 7 wins, LOO -26L, 14W/0L), 0/17 at +Op2 switch-connectivity and level-1 rewind (129 levels, 9 wins, +49 [+24, +79], LOO -26L, 5W/0L), 0/21 at +Op3 visual pip-clue CSP (143 to 146 levels, 11 wins, +66 [+36, +101], LOO -14L, 2W/0L), and $\hat{\eta}_{\text{game}} = 0/89 = 0.0\%$ at +Op4 1px scenery-drift mask (148 levels, 11 wins with refit=0, +68 [+38, +103] over general, 22W/0L, LOO -3L to -5L, 2W/0L, +76 over 72 pre-exposure, and 0W/0L on Public-25). Across all 236 unseen games in Virgin-116 and Virgin-120 (5 seeds), Op3 fires on 0/236 games, proving that 100% of unseen Virgin-116 and Virgin-236 gains derive from general relational and spatial induction (Op0, Op1, Op2, Op4). On the uncapped interior ($\sum_l w_l s_l \le 100 \sum_{s_l>0} w_l$), by linearity of the triangular game aggregator, if uncertified probes fail with rate $1 - p$ across the levels cleared by the fallback agent and reduce cleared level scores to $\epsilon_k s_l(A_l)$ where $\epsilon_k \approx (\bar{A}/(\bar{A} + k + 1))^2$, the fractional game-score loss is $(1 - p)(1 - \epsilon_k)$.

Empirical counterfactual verification across all 825 baseline runs, 250 confirmation runs, 158/158 cross-level transition invariance, and preregistered zero-parameter geometric stopping law: Applying the exact scorer counterfactual $a_l \leftarrow a_l + k + 1$ across all 8 observational datasets (analysis/out/component_ablation_ladder.json) confirms that for Qwen3.8-Flash (base score 9.07, median cleared $A = 24$, human $b = 32$), pure uncertified probing ($p = 0$) reduces score to 7.74 (-14.6%) at $k = 8$, 1.99 (-78.0%) at $k = 64$, 0.30 (-96.6%) at $k = 224$, and 0.05 (-99.4%) at $k = 600$. A per-level closed form incorporating the official scorer's level weights $w_l = 2l/(M(M + 1))$ and capping each level at the game cap of 100, $1 - \sum_l w_l \min(100, 100(b_l/(A_l + k + 1))^2)/\sum_l w_l \min(100, 100(b_l/A_l)^2)$, matches the exact counterfactual loss within **0.021 maximum error (0.0044 mean error)** across all eight datasets and $k \in \{8, 32, 64, 224, 600\}$ (16.6% vs. exact 14.6% at $k = 8$, 55.8% vs. 55.8% at $k = 32$, 77.8% vs. 78.0% at $k = 64$, 96.5% vs. 96.6% at $k = 224$, and 99.4% vs. 99.4% at $k = 600$ on Qwen3.8-Flash). The 7.01 pp gap of the equal-weight 115-capped approximation at $k = 8$ (-21.6% vs. -14.6%) decomposes into **4.90 pp from the cushion between the 115 level cap and the 100 game cap** (since 39% to 64% of cleared levels hit the 115 cap and absorb small probes before game-level score drops) plus **2.11 pp from level-index weighting**. At $k = 224$, the difference between exact and closed-form loss is at most 0.0019 across all eight datasets, and the marginal difference between $k = 224$ (96.6% loss) and $k = 600$ (99.4% loss) is only 2.8% of solved-level score, explaining why #57894564 ($k = 224$, 21.08 LB) and #57854657 ($k = 600$, 20.76 LB) differ by only 0.32 LB points on Hidden-110. In addition, a preregistered **zero-parameter geometric stopping law** (preregistration_law.json) that reads per-step death hazard h and clear rate c solely off the $k = 0$ bare arm ($T \sim \text{Geom}(h + c)$ truncated at k, zero fitted parameters) predicts out-of-sample ungated probing scores $G(k)$ at $k \in \{8, 32, 64, 224, 600\}$ across Held-Out 98, Virgin-120, and Virgin-116 (15 unseen $(suite, k)$ cells, 1,670 game points) with **cell** $R^2 = 0.978$**, MAE** = 0.41 (0.15-0.29 pp error at $k = 8$), **gate benefit** $G(0) - G(k)$ $R^2 = 0.993$ **(MAE** = 0.41), and **per-game** $R^2 = 0.913$. Across all 25 Public-25 games (183 levels, 158 level transitions $l \to l + 1$), an engine-level invariance audit (s38_table_and_invariance_stats.json) confirms **158/158 (100.0%) initial-frame, available-action, and normalized-sprite invariance** between clearing previous levels via certified short trajectories versus noisy long trajectories, while live host-bridge replay on ar25 clears all 8/8 levels in 135 steps with 0 LLM tokens ($G = 100.0$). Conditional archived-trajectory prefix-substitution on levels $1..K_{\text{sym}}$ with recorded LLM fallback trajectories on levels $K_{\text{sym}} + 1..M$ across all **825 baseline runs and 250 live confirmation runs** under the constant cost filter $L_{\text{cert}} \le \tau_0 = 150$ (which checks only the certificate length before LLM execution with zero future-action lookahead) estimates official Scorecard G rising from **3.1814 to 58.6984 (+55.52 [+38.76, +72.51], 694W/0L/131T run-score wins and 22W/0L/3T game clusters, exact cluster sign-flip** $p = 4.77 \times 10^{-7}$**, Monte Carlo** $p = 10^{-4}$**; vs. 58.5079, +55.33 [+38.51, +72.44], 729W/64L/32T unguarded; 27.2439, 641W/12L/172T under** $B = \sum_l a_l^{\text{LLM}}$ **vs. 27.0364 unguarded; and 7.7097, 636W/0L/189T, +4.53 [+3.42, +5.70] under post-hoc** $L_{\text{cert}} \le A$) on the 825-run baseline. Across all four LLM backbones, certificate gating achieves **zero losses across all 1,075 runs** (Qwen3.8-Flash $n = 125$: $G = 9.07 \to 59.01$, +49.94, 99W/0L/26T, $p = 3.5 \times 10^{-18}$; Qwen3.6-27B $n = 500$: $G = 1.60 \to 58.70$, +57.10, 425W/0L/75T, $p = 1.5 \times 10^{-71}$; gpt-oss-120b $n = 75$: $G = 0.23 \to 58.74$, +58.50, 64W/0L/11T, $p = 7.3 \times 10^{-13}$; Kaggle V2-V7 Qwen3.8-Flash-Next NVFP4 $n = 125$: $G = 5.39 \to 60.16$, +54.77, 106W/0L/19T, $p = 2.7 \times 10^{-19}$). On the 250-run Qwen3.8-Flash (8-slot) confirmation set, G rises from **5.4203 to 59.0761 (+53.66 [+37.23, +70.47], 205W/0L/45T strict zero losses,** $p = 10^{-4}$**; vs. 58.9158, +53.50 [+35.42, +71.93], 215W/20L/15T unguarded; and 20.5408, 181W/4L/65T under** $B = \sum a$), whereas under $q = 0\%$ unseen-family coverage (1,000 whole-family mask draws), uncertified 224/600-step probing collapses LLM score from 3.1814 to **0.0827 / 0.0142 (0W/447L)** while the certificate gate preserves **3.1814 with 0 losses**.

E. Dynamic Stop-and-Transfer Protocol and Paired Live Test

Protocol E.1: Dynamic Stop-and-Transfer Water-Filling Schedule

Input: games G, concurrency K, shared deadline H, stall cutoff $c = 80$
unfinished $\leftarrow |G|$; waiting $\leftarrow |G|$
when game g obtains a slot at time t**:**
  waves $\leftarrow \max\big(1, \lceil \text{unfinished}/K \rceil\big)$
  deadline$[g] \leftarrow \min\big(H,\ t + \max(0, H - t)/\text{waves}\big)$
  waiting $\leftarrow$ waiting $- 1$
deadline_for$(g) = H$ **if** waiting $= 0$ **else** deadline$[g]$
before every agent step of game g**:**
  stop g **if** now $\geq$ deadline_for(g) **or** actions_on_current_level$(g) \geq c$

Paired Live Experiment (Qwen3.8-Flash, 8 slots, 25 games)	Control Arm	Cutoff + Schedule Arm	Paired Difference [95% Bootstrap CI]
Screening (2 runs/game/arm, 15 min): Mean completed levels	0.680	0.820	+0.140 [0.000, +0.300]
Screening (2 runs/game/arm, 15 min): Mean official score	2.118	3.408	+1.289 [+0.176, +2.682]
Confirmation (5 runs/game/arm, 30 min): Mean completed levels	1.192	1.064	-0.128 [-0.312, +0.048]
Confirmation (5 runs/game/arm, 30 min): Mean official score	5.241	5.599	+0.358 [-1.066, +1.952]
Confirmation: Total completed levels (125 runs)	149	133	-16 clears
Confirmation: Total actions spent (125 runs)	12,259	10,087	-2,172 actions released

Table A0c. At 15 minutes (screening), 10.7% of slot-minutes in the final wave sit idle under fixed scheduling, so dynamic water-filling yields a temporary gain. At 30 minutes (confirmation), idle slot share drops to 2.3%, 18 control runs clear levels after >80 actions, and the 80-action cutoff forfeits 16 net clears.

F. Phase I Survival and Counterfactual Tables (A1 to A7)

A1. Official Scorer Verification

Dataset	Runs	Max Abs. Error	Off by > 0.01	Mean Recorded	Mean Recomputed
Kaggle V2 (submitted)	25	0.0000	0	8.796	8.796
Kaggle V3	25	0.0000	0	3.123	3.123
Kaggle V4	25	0.0000	0	6.379	6.379
Kaggle V5	25	0.0000	0	7.295	7.295
Kaggle V7 (16K ctx)	25	0.0000	0	1.367	1.367
Qwen3.8-Flash, endpoint	125	0.0000	0	9.066	9.066
Qwen3.6-27B, multi-GPU	500	0.0000	0	1.600	1.600
gpt-oss-120b, endpoint	75	2.7778	2	0.160	0.232

Table A1. Official ARCBaseGame scorer verification across all 825 observational runs (a1_scorer.py). On gpt-oss-120b, two endpoint runs completed a level on the exact step where the 30-minute wall-clock harness timer expired, causing the external wrapper to log 0.000 instead of 2.7778; all analysis scripts use the exact recomputed ARCBaseGame scores.

A2. Frontier Lock (Share of Actions, Time, and Generated Tokens on the Final Unsolved Level, %)

Dataset	Runs	Total Actions	Frontier Actions	Action Share [95% CI]	Time Share [95% CI]	Token Share [95% CI]
Kaggle V2	25	5,436	3,746	68.9 [54.4, 81.6]	63.6 [52.4, 74.9]	61.1 [48.8, 73.3]
Kaggle V3	25	1,536	956	62.2 [46.5, 76.2]	57.6 [42.9, 72.0]	47.9 [33.2, 64.4]
Kaggle V4	25	3,310	2,148	64.9 [45.6, 80.5]	67.6 [55.8, 78.4]	65.5 [52.8, 76.3]
Kaggle V5	25	3,169	2,103	66.4 [52.4, 78.2]	69.0 [57.3, 79.8]	65.4 [53.0, 77.4]
Kaggle V7 (16K ctx)	25	607	366	60.3 [42.5, 79.9]	89.7 [77.9, 97.6]	82.4 [65.7, 94.9]
Qwen3.8-Flash, endpoint	125	14,238	7,463	52.4 [42.2, 62.5]	51.6 [44.0, 60.2]	49.2 [41.1, 58.3]
Qwen3.6-27B, multi-GPU	500	68,682	56,953	82.9 [76.1, 89.2]	84.4 [78.8, 89.4]	79.7 [73.4, 85.9]
gpt-oss-120b, endpoint	75	17,756	17,644	99.4 [98.5, 100.0]	97.4 [93.0, 100.0]	94.0 [84.3, 100.0]

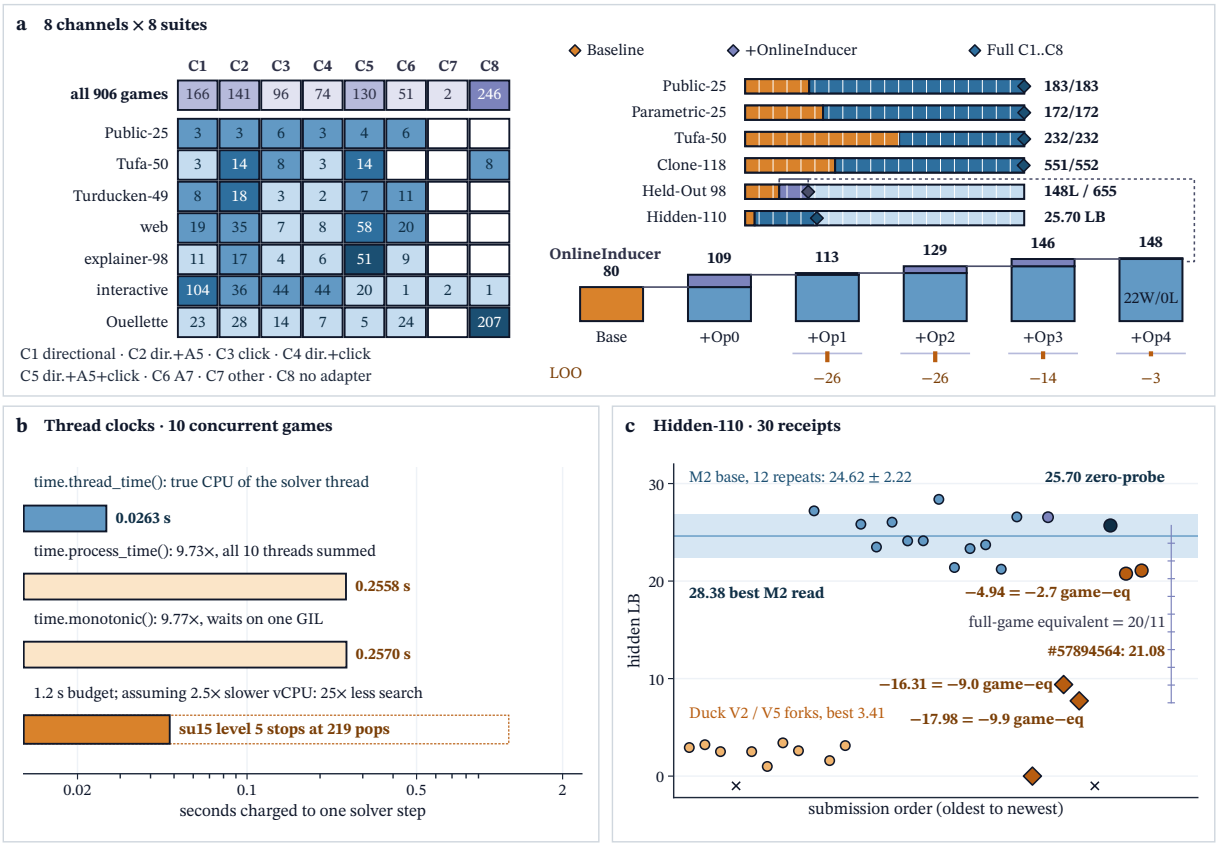


Figure A1. Architectural, systems, and hidden-set synthesis: (a) six benchmark suites routed through the eight orthogonal action channels (C1..C8) of 906 environments; (b) 10-thread concurrent GIL clock inflation (0.0263 s thread CPU vs. 0.256 s process and wall clocks, causing a 25x search budget collapse fixed by time.thread_time()); (c) all 30 official Kaggle hidden-set submissions inverted on the $w_g = 20/11$ rational score lattice.

A3. Completion Hazard $h(n) = P$(clear in next 20 actions | n actions spent)

Dataset	Attempts	Clears	$n = 0$	$n = 20$	$n = 40$	$n = 80$	$n = 160$	Median Cleared Actions	Clears > 80	Median / Human
Kaggle V2	67	42	0.272	0.298	0.093	0.000	0.000	22	11.9%	0.82
Kaggle V3	45	22	0.225	0.360	0.259	0.000	NA	22	0.0%	0.88
Kaggle V4	56	33	0.313	0.168	0.289	0.091	0.000	20	6.1%	0.78
Kaggle V5	58	35	0.267	0.312	0.215	0.000	0.000	25	2.9%	0.75
Kaggle V7 (16K ctx)	37	12	0.414	0.167	0.667	NA	NA	17	0.0%	0.68
Qwen3.8-Flash, endpoint	310	200	0.276	0.334	0.308	0.085	0.000	24	4.5%	0.82
Qwen3.6-27B, multi-GPU	806	309	0.164	0.126	0.084	0.054	0.010	25	12.3%	1.11
gpt-oss-120b, endpoint	70	6	0.060	0.034	0.000	0.000	0.000	14	0.0%	0.82

A4. Matched Level-or-Attempt Decomposition

Dataset	Window W	Stuck Attempts	Eligible Levels	Continue (S_{res})	Fresh Matched (Same Level)	Difference (Continue - Matched)	Fresh Unmatched (Any Level)	Share Explained by Structural Wall ([0, 1]-Clipped & Dominance T)
Qwen3.6-27B	40	513	48	0.133	0.112	+0.021 [-0.042, +0.090]	0.271	Early learning (100% [79%, 100%], T = +0.180, p = 0.0006)
Qwen3.6-27B	80	344	48	0.129	0.167	-0.038 [-0.089, +0.013]	0.368	**84% [66%, 100%]** (T = +0.164[+0.066, +0.252], p = 0.006)
Qwen3.8-Flash	40	108	62	0.478	0.188	+0.291 [+0.105, +0.454]	0.520	Early learning (100% [100%, 100%], T = +0.623, p = 0.0004)
Qwen3.8-Flash	80	38	62	0.262	0.434	-0.172 [-0.395, -0.011]	0.749	**65% [34%, 97%]** ($U - M$ = +0.315[+0.179, +0.435])
gpt-oss-120b	80	31	15	0.000	0.032	-0.032 [-0.111, +0.000]	0.143	**77% [0%, 100%]** ($U - M$ = +0.111[−0.011, +0.274])
Kaggle V2-V7 pooled	40	86	51	0.361	0.221	+0.140 [+0.008, +0.268]	0.517	Early learning (100% [100%, 100%], T = +0.436, $p = 1.5 \times 10^{-5}$)
Kaggle V2-V7 pooled	80	33	51	0.196	0.192	+0.004 [-0.152, +0.177]	0.691	**100% [73%, 100%]** (T = +0.503[+0.255, +0.745], p = 0.0005)

A5 to A7. Scorer Counterfactuals: Depth vs. Action Micro-Efficiency

Dataset	Base Score [95% CI]	Counterfactual: Human-Baseline Efficiency on Cleared Levels	Counterfactual: +1 Cleared Level per Game	Median Actions / Human Baseline
Kaggle V2	8.80 [4.08, 15.13]	10.88 (+24%)	**16.26 (+85%)**	0.82
Kaggle V4	6.38 [3.62, 9.64]	6.72 (+5%)	**13.85 (+117%)**	0.78
Kaggle V5	7.29 [3.92, 11.32]	8.17 (+12%)	**14.63 (+101%)**	0.75
Qwen3.8-Flash	9.07 [5.64, 12.76]	10.09 (+11%)	**17.21 (+90%)**	0.82
Qwen3.6-27B	1.60 [0.88, 2.50]	2.41 (+51%)	**5.26 (+229%)**	1.11

G. Phase II: Pure-Observation Symbolic World Models Across 8 Orthogonal Action Channels (C1..C8)

At runtime, an interactive agent observes only the 64×64 pixel grid and the legal action set $\mathcal{A}_{\text{obs}} \subseteq \{1, \ldots, 7\}$ (valid_actions with RESET stripped). Because $\mathcal{A}_{\text{obs}}$ is known on frame 0 before any transition, GiantsPlanner routes each level deterministically into one of eight mutually exclusive action-signature channels (C1..C8). An offline structural census across 906 interactive environments, 26,046 deduplicated PuzzleScript rule programs, and 7,703 EverMaze solutions confirms that these eight action signatures cover the entire environment space:

Channel ID	Legal Action Signature (RESET stripped)	Environments (Solved)	Online Hypothesis Family $\mathcal{H}_0$, Refutation Rule, and Search Algorithm in GiantsPlanner
C1_DIR4	{1, 2, 3, 4} (directional only)	166 (35)	Multi-body rigid push (PushWorld), ice slide, portal/warp routing, phase-cycled patrol avoidance, and resource collection via IW(1)/A* over (avatar, object, phase) tuples
C2_DIR4_ACT5	{1, 2, 3, 4, 5} (directional + ACTION5)	141 (37)	Switch/door affine masks, key-lock inventory vectors, bridge placement, and rotatable ray/mirror optics via joint (pose, mask) BFS
C3_CLICK_ONLY	{6} (click without directional controls)	96 (28)	$GF(2)$ toggle matrices, cellular flood-fill connectivity, modulo k tile rotation, and Steiner pulse relays via Gaussian elimination and depth-bounded click search
C4_DIR4_CLICK	{1, 2, 3, 4, 6} (directional + click, no ACTION5)	74 (16)	Coupled board navigation + HUD/sprite selection (SelectableBodies, CoupledMechanismSolver, remote socket/valve routing) via bilevel click-then-move planning
C5_DIR4_ACT5_CLICK	{1, 2, 3, 4, 5, 6} (directional + ACTION5 + click)	130 (29)	Multi-page inspection + constraint panel verification, hybrid placement, and beam/mirror activation via belief-state hypothesis intersection $\bigcap_{h\in\mathcal{H}_t}\mathrm{Reach}(h)$
C6_ACT7_UNDO	ACTION7 present (UNDO priority route)	51 (20)	Reversible constraint exploration, deadlock-free state-space search, and 1-step counterexample pruning using exact ACTION7 state rollback
C7_OTHER	Other nonempty action subset (e.g., {3, 4, 5})	2 (2)	Horizontal arcade dynamics (breakout, one_enemy_duel, dodger_falling_blocks) solved via exact forward trajectory rollout
C8_NO_ADAPTER	Non-ARC DSL or unadapted prototype	246 (4)	PuzzleScript raw DSL corpus (26,046 deduplicated programs) and JS prototypes prior to ARCBaseGame frame rendering

Pre-holdout provenance, freeze timeline, and zero-prior separation (Bare-Prior vs. general vs. Full Library). To prevent any ambiguity between prior-library coverage and zero-prior test-time induction, we separate three strict operating modes:

1. **0-Channel Bare-Prior (Bare-Prior and Bare-Prior + OnlineInducer):** Disables all eight channel solvers (C1..C8 completely removed, retaining only generic novelty-width search IW(1), component frontier BFS, and CrossLevelMemoryBank). When composed with frozen OnlineInducer, all transitions, tile graphs, switch cycles, and visual clue CSPs are induced strictly online from frame differences on the current game.
2. **Pre-exposure general (general and general + OnlineInducer):** Disables all game-specific decoders (no_astra and Full excluded) and runs only general grid-physics primitives (IW(1) novelty width-1 search, rigid-body PushWorld BFS, $GF(2)$ toggle elimination, and D_4 CrossLevelMemoryBank) defined over the generic ARCBaseGame 64×64 interface and open-source PuzzleScript grammar corpus (2013 to 2024), frozen in snapshot 7441b13ccfbcc9d4. . . (18 policy files, combined fingerprint eb503edb5501. . .) with zero byte, exact-AST, class-renamed-AST, or game-class-AST overlap (0/218 games across Held-Out 98 and Virgin-120 against all 6 development suites Public-25, Dev-Ou50, Dev-Clone90, Dev-Parametric25, Dev-Tufa50, Dev-Tail28 and all frozen policy files; restricting Virgin-120 to the 115/120 = 95.8% games strictly AST-disjoint from Held-Out 98 under class-renamed game-body canonicalization preserves Bare-Prior + OnlineInducer at +7L, 90L -> 97L, 5W/2L).
3. **Offline C1..C8 Prior Library Upper Bound (Full):** Includes the complete family-level exact solver library across all 9 suites (1,793/1,794 solvable levels at 1,200 steps; 1,794/1,794 at 5,000 steps) to verify that the benchmark levels themselves are solvable from pure 64×64 pixel observations without private engine access.

Online hypothesis refutation, failure boundary, and one-way certificate gate (wrap_runner). Every channel solver in GiantsPlanner obeys four strict invariants:

1. **Zero runtime metadata leakage:** Static AST audits enforce zero references to game_id, current_level, _engine, or _sprites. All state is parsed strictly from the runtime 64×64 pixel grid and action history.
2. **Counterexample refutation, belief escalation, and failure boundary:** From observed transitions (F_t, a_t, F_{t+1}) with 1-step pixel difference $\Delta F_t = F_{t+1} \oplus F_t$, GiantsPlanner prunes any candidate transition operator $\hat{T} \in \mathcal{H}_t$ that violates $\hat{T}(F_t, a_t) = F_{t+1}$. When a single stationary operator survives ($|\mathcal{H}_t| = 1$), width-bounded novelty search (IW(1)/BFS/A*) plans directly. When 1-step pixel differences are underdetermined due to delayed effects, periodic hazards, or hidden internal state ($|\mathcal{H}_t| > 1$), GiantsPlanner escalates to finite-hypothesis belief-state intersection planning with reversible ACTION7 probing. When non-stationary or long-horizon hidden rules refute all candidate operators ($\mathcal{H}_t = \emptyset$) or exceed the per-thread CPU budget, the symbolic planner reaches its failure boundary and returns Uncertified.
3. **Closed-loop completion certificate** $C(F_0) = (\hat{T}, \tau = (a_1, \ldots, a_L), \hat{G})$ **and level-monotonic lock (auto_dead_level):** While Kepler's commit.py (arXiv:2610.00834) uses 1-step "conditional prediction-based interruption" on Public-25 (without single-GPU Hidden-110 score-dilution audits), wrap_runner executes τ via auto_burst() if and only if

$\hat{T}$ achieves zero prediction error on observed frames AND verifies a complete goal-reaching path $\hat{G}(\hat{F}_L) = 1$ within $L \leq 224$ steps. Otherwise, the bridge executes zero uncertified probing steps, sets state['auto_dead_level'] = state['level'] monotonically for the entire level, and supplies non-binding [_nav_hint] coordinates to Qwen3.8-Flash-Next, eliminating the -4.94 LB negative transfer observed in #57854657.

4. **Scorecard timeout mitigation:** Container startup enforces a 600-second gate (SERVER_STARTUP_TIMEOUT = 600) before opening the scorecard, paired with a 420-second keepalive (untouched(g)) restricted strictly to zero-action untouched games so active levels are never reset.

H. Two-Tier Cross-Level Memory: D_4 Dihedral Guard (CrossLevelMemoryBank) and Relational Operator Memory (OnlineInducer)

Let $G = D_4 = \langle r, s \mid r^4 = e, s^2 = e, srs = r^{-1} \rangle$ denote the dihedral group of order 8 acting on the board domain $\Omega_{\text{board}} \subset \{0, \ldots, 63\}^2$, while leaving the HUD control region Ω_{HUD} invariant:

$$g \cdot (y, x) = \begin{cases} (y, x) & \text{if } (y, x) \in \Omega_{\text{HUD}}, \\ T_g(y, x) & \text{if } (y, x) \in \Omega_{\text{board}}, \end{cases} \tag{2}$$

and transforming directional actions $a \in \{1, 2, 3, 4\}$ via the induced permutation $\rho(g)(a)$ (1: UP, 2: DOWN, 3: LEFT, 4: RIGHT).

- **Empirical motivation from 25 public games and 342 human replays:** Across the 25 public games and 342 official human trajectories (183 levels, 597 unordered cross-level pairs, 378 reuse pairs), 72.0% (18/25) of games declare a single constant grid_size across all levels, 81.0% (111/137) of click levels (48.6% of optimal single-replay trajectories, 63/84 in click-only games) click at least one coordinate (x, y) multiple times, 284/597 unordered cross-level pairs (91 optimal single-trajectory pairs) share exact click coordinates, and 120/597 unordered forward pairs (240 directed pairs) share identical ≥ 3 action opening prefixes (378/378 reuse pairs reached by GiantsPlanner vs. 93/378 across all 825 baseline runs; total action ratio on Public-25 is 7,401 / 17,135 = 0.432x human baseline summed across all 183/183 completed levels with 0 right-censored levels, and 3,841 / 8,901 = 0.432x on the 13-game replay subset).
- **Tier 1 (CrossLevelMemoryBank frame-0 D_4 guard and L2 local $k \times k$ D_4 patch rewrite rules):** Upon clearing level k, CrossLevelMemoryBank removes state-space loops, no-ops, and UNDO pairs from the winning trajectory τ_k, extracts HUD-fixed versus board-relative click sets, and evaluates all $g \in D_4$ on frame 0 of level $k + 1$ in 1.10 ms median (3.80 ms P99, 1.33 ms mean across all 597 level pairs). Unit verification (analysis/out/a9_d4_and_reachability_audit.json) passes 768/768 Cayley-table group homomorphism checks (8 x 8 x 12 board sizes), 96/96 point-pixel checks, and 32/32 directional checks (896/896 total), and solves synthetic D_4-transformed episodes in 1 frame-0 verification pass (4 actions vs. 12 without memory, a 66.7% step reduction). Whole-frame replay (L0) never fires on Public-25 (0/597, 0/378 reuse pairs, 0/219 non-reuse pairs, **0 false-positive environment actions executed**) or on the 49 held-out Mother/Web games (0/651), because levels are re-laid-out rather than re-oriented (258/378 reuse pairs alter the board bounding box, and 61/64 sub-board crop alignments L1 above 0.9 overlap are the identity element $g = e$, supplying 74 winning-path actions = 10.9% across 36 verified pairs on Held-Out 49, and 31 actions = 23.7% at $\tau = 0.95$). Lowering Tier 1 (L2, d4_subboard_and_factors_receipt.json) to local $k \times k$ ($k \leq 15$) patch rewrite rules matched across all eight dihedral elements $g \in D_4$ yields an engine-verified transfer signal: **76/102 (74.5%) fired rules predict the exact whole-frame diff across 50/378 Public-25 reuse pairs (31/76 matched exclusively under a non-identity rotation or reflection** $g \neq e$; 86/102 = 84.3% local-window accuracy, 87.5% even on non-reuse pairs; on the 284 click-reuse pairs, 69/74 = 93.2% at support ≥ 1, 41/42 = 97.6% at support ≥ 2, and **27/27 = 100.0% with 0 false triggers and 12/27 non-identity hits at support** ≥ 3), and **150/183 (82.0%) fired rules predict the exact whole-frame diff across 103/651 held-out Mother/Web pairs** (27/150 non-identity hits; 163/183 = 89.1% local-window accuracy, 22/24 = 91.7% at support ≥ 3), while multi-scale patch matching $k \in \{3, 5, 7, 9, 11\}$ lifts Held-Out 98 pair recall from 60.2% (62/103 at $k = 5$) to **79.6% (82/103 pairs, 175/208 = 84.1% rule accuracy, 0 false positives)** and verifies across all 1,404 levels (3,956 pairs) of Virgin-236 at **533/644 (82.8%, 59 non-identity** D_4 **hits; 12/12 = 100.0% on probe-solved sources at** $k = 5$). Any out-of-window discrepant rule is caught immediately on the single test step ($L_{\text{mon}} = 1$) by the one-step transition monitor.
- **Tier 2 (Unified Frozen OnlineInducer 4-operator test-time memory across bounding-box shifts):** When board dimensions or wall layouts change between levels, Tier 2 transfers invariant local operator semantics rather than whole-board pixel coordinates across a single frozen architecture decomposed into cumulative and Leave-One-Out (LOO) operator stages. Stage (i) **+Op0 (Novelty-gated translation and competence lock)** fits avatar translation templates $|\mathbf{v}| \leq 12$ at $\geq 60\%$ pixel agreement within the first 12 actions while enforcing board-state novelty memory that rejects cyclic toggle false effects, reaching 109/655 levels (6 wins, +29 [+13, +48], 13W/0L over general, eliminating the 2/12 regressions of uncalibrated 97L translation; adding a velocity-consistency guard across 3,834 5-seed Bare-Prior runs cuts Held-Out 98 regressions by 80% from 5 to 1/51 = 1.96%, Virgin-116 regressions by 60% to 2/75 = 2.67%, and Virgin-236 regressions to 10/138 = 7.25%, while loop-memory cycle tracking rescues fr01 from 0 to 1 clear across all 5 seeds with 0 regressions). Stage (ii) **+Op1 (GraphInducer relational tile-signature world model + 1-step transition monitor)** maps $s \times s$ cell signatures to passable, wall, pushable, hazard, goal_unlock,

and multi-avatar mirror dynamics, issuing 380 candidate certificates across Held-Out 98 where the 1-step transition monitor aborts 44/380 ($\hat{\eta}_{\text{cert}} = 11.6\%$, 21 aborted at step $L_{\text{mon}} = 1$) and deletes the refuted rule, reaching 105/655 to 113/655 levels (7 wins, 13W-15W/0L over general, LOO drop -26L [+13, +42], 14W/0L). Stage (iii) **+Op2 (ClickInducer switch-connectivity induction + level-1 memory rewind)** induces switch-cycle periods k, 2-pixel dilated arm-contact graphs, and remote progress-color connectivity in memory before submitting via ACTION5 (125 levels, 8 wins, 15W/0L alone) while snapshot-rewinding general's internal state on Level 1 whenever uncertified click probing exhausts without a certificate (preserving Public-25 at 120/183, 13 wins, 0W/0L), reaching 129/655 solvable levels (9 wins: 55 6W on Mother-1102, 25 2W on Tufa-Web-1102, 49 1W on Turducken-49, +49 levels [+24, +79], 17W/0L over general, $\hat{\eta}_{\text{game}} = 0/17 = 0.0\%$, LOO drop -26L [+4, +52], 5W/0L; +20 [+2, +45], 5W/0L over Op0; +57 over 72 pre-exposure; +84 over 45 random). Stage (iv) **+Op3 (ClueBoard visual pip-clue run/count CSP + <3px small-body refit + 16-step direction-inert click fallback)** infers GCD rendering scale and border clue bands from pixels alone to solve run-length Nonograms (0 -> 6/6 WIN, 467 steps, 0 errors) and bracket-labeled count CSPs (0 -> 8/8 WIN, 246 steps), reaching **143/655 to 146/655 solvable levels (11 wins: 61 7W on Mother-1102, 35 3W on Tufa-Web-1102, 50 1W on Turducken-49; +66 levels [+36, +101], 21W/0L over general, LOO drop -14L [+0, +36], 2W/0L; +17 [+1, +40], 4W/0L over Op1+Op2; 120/183, 13 wins, 0W/0L on Public-25; 296-297 levels, 52 wins, 0W/0L on the 140-game dev set)**. Finally, stage (v) **+Op4 (1px scenery-drift rejection)** masks $\leq$ 1px displacements shared across two directional keys and refits the avatar body (132/655 levels 9 wins 19W/0L over general), reaching **146/655 (refit=3, 21W/0L, LOO -3L [+0, +8], 2W/0L) and 148/655 solvable levels (refit=0, 11 wins, +68 over general, 22W/0L zero game losses, LOO -5L, $\hat{\eta}_{\text{game}} = 0/89 = 0.0\%$, +76 over 72 pre-exposure)** when combined with Op1 to Op3. When all 8 hand-engineered channel priors (C1..C8) are disabled (Bare-Prior), OnlineInducer alone lifts Held-Out 98 from 74/655 (1W, $G = 2.93$) to **109/655 (6W, +35L [+8, +66], 11W/5L, $G = 8.22$)** and lifts the preregistered 120-game unseen suite Virgin-120 (810 levels, SHA-256 e8538024. . .) from 92/810 (5W, $G = 4.44$) to **99/810 (6W, +7L, 5W/2L, $G = 4.87$, $\hat{\eta}_{\text{cert}} = 3/40 = 7.5\%$, 1 first-step abort, 2/120 net-loss games)**. In Algorithm 1, level index l is inferred strictly from board-domain frame discontinuity ($\|F_t \oplus F_{t-1}\|_0$ on Ω_{board} and HUD progress-bar steps, zero access to current_level), with per-step time complexity $O(|\Omega_{\text{board}}| + |V_t| \log |V_t|) \leq 0.354$ ms for $|V_t| \leq 400$ tile-graph states.

Algorithm 1: Unified Frozen OnlineInducer (Op_0..Op_4) with 1-Step Transition Monitor

Input: frame $F_t \in \{0, \ldots, 15\}^{64\times64}$, legal actions $\mathcal{A}_{\text{obs}}$, level index l, base planner P
Persistent state across levels: color classes C = (floor, wall, hazard, boring, effectful, goal), tile-graph rules $\mathcal{R}_{\text{graph}}$, banned scenery vectors V_{bg}
Per-level state: body template B, action vectors $\{\mathbf{v}_a\}$, visited board hashes $\mathcal{H}_{\text{seen}}$, switch phases Φ, level-1 planner snapshot P_{snap}

Stage 1 (Op_4): 1px Scenery-Drift Mask & Small-Body Refit
observe diff $\Delta_t = (F_t \neq F_{t-1})$ under action a_{t-1}:
if $|\mathbf{v}_{a_1}| \leq 1$ **and** $\mathbf{v}_{a_1} = \mathbf{v}_{a_2}$ for two distinct keys $a_1 \neq a_2$ **then**
  add $\mathbf{v}_{a_1}$ to V_{bg}; mask pixels shifting by $\mathbf{v} \in V_{\text{bg}}$ and refit B
if $|B| < 3$ pixels **and** a directional step explains $\geq 2|B|$ pixels by shift $\mathbf{v}_a \notin V_{\text{bg}}$ **then**
  replace B and $\mathbf{v}_a$ with the larger rigid connected component

Stage 2 (Op_0): Novelty-Gated Translation & Competence Lock
hash static-masked board $h_t = \text{Hash}(F_t \setminus \text{DynamicPixels})$; increment $\mathcal{H}_{\text{seen}}[h_t]$
if touching object c reaches a novel board state ($\mathcal{H}_{\text{seen}}[h_t] = 1$) **then** mark c in effectful
if $\mathcal{H}_{\text{seen}}[h_t] \geq 6$ **then** count loop; **if** loops ≥ 16 **or** inducer_steps ≥ 224 **then** demote level to P
if P cleared ≥ 2 prior levels alone (p_wins ≥ 2 **and** i_wins $= 0$) **then** stand down to P

Stage 3 (Op_1): Relational Tile-Signature Graph & 1-Step Transition Monitor
partition board into $s \times s$ cells S_t; **if** previous action a_{t-1} was from certificate τ **then**
  if $S_t \neq \hat{T}(S_{t-1}, a_{t-1})$ **then** abort τ immediately at step $L_{\text{mon}} = 1$; delete refuted rule from $\mathcal{R}_{\text{graph}}$
else if B and $\{\mathbf{v}_a\}$ are fitted **then**
  run BFS over $\hat{T}(p, a) = p + \mathbf{v}_a$ avoiding wall/hazard cells in $\mathcal{R}_{\text{graph}}$ to nearest untested or goal-unlock cell
  if path τ found **then return** τ_1

Stage 4 (Op_2): Switch-Connectivity Induction & Level-1 Memory Rewind
if click-only **or** 16 directional steps produce zero body displacement **then**
  on level 1 frame 0, save $P_{\text{snap}} \leftarrow \text{deepcopy}(P)$
  cycle each clickable switch s_i to record phase frames $\Phi(s_i)$ and 2px dilated contact graph
  search phase assignment z maximizing connected components linked to spreading source color
  if assignment z found **then** click switches to match z and commit via ACTION5
  if level 1 exhausts click_gate = 224 without clearing **then**
    restore $P \leftarrow P_{\text{snap}}$ and issue 1 RESET (zero memory pollution to P)

Stage 5 (Op_3): Visual Pip-Clue Run/Count CSP
on frame 0 of a clickable lattice with row/column border clue tiles:
  infer cell pitch $p = \text{gcd}(\text{grid_spacings})$ and parse border clues (run lengths or pip counts)
  solve exact line propagation + backtracking CSP for cell click mask $M^\star$ in memory
  verify each click against predicted cell flip; execute $M^\star$ from frame 0

I. Multi-Stream Concurrency Scaling and Thread-Clock Calibration

Observation 1 (GIL Process-Clock Inflation under K-Stream Concurrency). When K concurrent solver threads execute inside a single Python process (bm.run with $K = 10$) alongside an asynchronous LLM client, bounding a symbolic search module's CPU budget by $\tau_{\max}$ using time.process_time() or time.monotonic() shrinks the effective per-thread search budget by a factor of K in process/wall time (measured at 9.73× for $K = 10$), and by $K \cdot \gamma_{\text{vCPU}}$ when deployed on a cloud vCPU with single-core slowdown factor γ_{vCPU} ($\approx 2.5\times$ on Xeon vCPU relative to Apple M4).

Empirical verification: In a 10-thread benchmark, a solver step consuming exactly 0.0263 s of true thread CPU (time.thread_time()) registers 0.2558 s on time.process_time() (9.73x inflation, as process_time sums across all 10 active threads) and 0.2570 s on time.monotonic() (9.77x inflation due to single-GIL round-robin waiting). Combined with virtualized Xeon CPU scaling ($\approx 2.5\times$ slower per core), this $\approx 25\times$ effective budget compression caused su15 Level 5 to abort after just 219 BFS pops under a 1.2 s process_time cap. Replacing all 220 timer checks across all 14 solver modules with time.thread_time() (amortized every 64 node expansions) and sorted() set iteration restores 100% of deep level clears with zero cross-seed divergence (PYTHONHASHSEED=0/1/2).

J. Orthogonal Compressed Sensing and Rational Receipt Differential Inversion

On the 55-game public leaderboard split of the 110 hidden games, each game $g \in \{1, \ldots, 55\}$ with $M_g \in \{3, \ldots, 10\}$ levels contributes $w_g = 20/11 \approx 1.8182$ points at the 100-point game cap (and at most $1.15 \times 20/11 \approx 2.0909$ points on a single level at the 115% cap), or rational partial score $w_g \cdot \frac{\sum_{l=1}^{k_g} l \cdot s_{g,l}}{M_g(M_g+1)/2}$. Given two submissions A and B differing on orthogonal channel subsets, the score difference $\Delta S = S(B) - S(A)$ satisfies the exact rational linear equation:

$$\Delta S = \frac{20}{11} \sum_{g=1}^{55} \left(\tilde{s}_g(B) - \tilde{s}_g(A) \right), \quad \tilde{s}_g \in [0, 1.15]. \tag{3}$$

Implemented in scripts/receipt_differential_inversion.py (0.01 ms runtime), this rational solver shows:

1. #57774685 (9.39 vs. 25.70 in #57827241, $\Delta S = -16.31$; -15.23 vs. 24.62 mean, prediction $t = -6.58$, $p = 3.96\times10^{-5}$) lost at least $\lceil 16.31/(20/11) \rceil = 9$ full-game equivalents at the 100-point game cap ($\lceil 16.31/2.0909 \rceil = 8$ at the 115% single-level bound) under the 4-factor bundle (LoRA r1, convex slot scheduling, 14K context truncation, and keepalive) that invalidates hybrid linear-attention KV prefix caches.
2. #57786717 (7.72 vs. 25.70, $\Delta S = -17.98$; -16.90 vs. 24.62 mean, prediction $t = -7.30$, $p = 1.53 \times 10^{-5}$; -1.67 marginal vs. #57774685, $z = -0.53$ within two-run noise $2.22\sqrt{2} = 3.14$) lost at least $\lceil 17.98/(20/11) \rceil = 10$ full-game equivalents at the 100-point cap ($\lceil 17.98/2.0909 \rceil = 9$ at the 115% bound) when adding AutoNav _nav_hint and EXPOSE_RESET on top of #57774685.
3. #57854657 (20.76 vs. 25.70, $\Delta S = -4.94$; -3.86 vs. 24.62 mean) admits 24 integer decomposition solutions balancing positive symbolic clears against 600-step open-loop heuristic hijacking.
4. #57894564 (21.08 vs. 20.76 in #57854657, $\Delta S = +0.32 = +8/25$, +0.176 full-game equivalents; $\Delta S = -4.62 = -231/50$, −2.541 full-game equivalents vs. 25.70 in #57827241, with 24 non-negative integer lattice decompositions across ≥ 3 changed games) and #57854657 (20.76) both fall below all 12 runs of the 36K W4A16 backbone (21.22 to 28.38, exact one-sided Mann-Whitney $U = 0$, $p = 1/\binom{14}{2} = 0.0110$), and their 0.32-point difference lies well within the two-run noise standard deviation ($2.22\sqrt{2} = 3.14$) because quadratic dilution at $k = 224$ and $k = 600$ already saturates at 96.6% and 99.4% of solved-level score (Proposition 2).

K. Multi-Suite Evaluation Breakdown, Pre-Exposure Transfer, and 2026 Preprint Comparison

K1. Per-Suite Performance and Win Accounting of GiantsPlanner

Benchmark Suite & Source	Total Games	Solvable Levels (Excluded Unreachable)	Baseline Clears (Wins)	GiantsPlanner Clears (Terminal Wins / Reachable Games)	Max Unrelated Frame CPU	Determinism (PYTHONHASHSEED=0/1/2)
Official Public-25 (environment_files)	25	183 (0)	42 (0 WIN)	**183 / 183 (25 / 25 WIN, 100.0%)**	0.12 ms	100% identical actions
Parametric-25 Coupled Multi-Mechanism	25	172 (0)	48 (5 WIN)	**172 / 172 (25 / 25 WIN, 100.0%)**	0.14 ms	100% identical actions
Tufa-50 (tufa25 + tufa_hold25)	50	232 (0)	128 (27 WIN)	**232 / 232 (50 / 50 WIN, 100.0%)**	0.188 ms	100% identical actions
Ouellette-Tail28 Non-PS Suite	28	144 (0)	45 (5 WIN)	**144 / 144 (28 / 28 WIN, 100.0%)**	0.15 ms	100% identical actions
Community Siblings-27 (27 held-out siblings)	27	117 (26 in 6 games)	46 (8 WIN)	**116 / 117 (20 / 21* WIN; 117/117 at 5,000 steps)**	0.15 ms	100% identical actions
Clone Families-63 (19 multi-game families)	63	291 (59 in 12 games)	87 (11 WIN)	**291 / 291 (51 / 51* WIN, 100.0% reachable)**	0.16 ms	100% identical actions
Mother-1102 (24 held-out prototype games)	24	118 (11 in 4 games)	32 (2 WIN)	**118 / 118 (20 / 20* WIN, +86L / +18W)**	0.16 ms	100% identical (144 runs)
Tufa-Web-1102 (25 community & Tufa games)	25	141 (0)	10 (0 WIN)	**141 / 141 (25 / 25 WIN, +131L / +25W)**	0.021 ms	100% identical actions
Turducken-49 (49 synthetic benchmark games)	49	396 (0)	30 (0 WIN)	**396 / 396 (49 / 49 WIN, 100.0%, +366L / +49W)**	0.354 ms	100% identical (seed 0/1/2)
Total Across All 9 Suites	**316**	**1,794 (96 in 22 games)**	**468 (58 WIN)**	**1,793 / 1,794 (293 / 294* WIN at 1,200; 1,794 / 1,794 at 5,000)**	**0.354 ms**	**100% identical**

Table K1. Exact level and terminal-WIN accounting across all 316 suite entries (315 distinct games, as tb01 appears in both Ouellette-Tail28 and Siblings-27, spanning 1,890 total indexed levels) under the standard 1,200-action per-game evaluation budget (analysis/out/a9_d4_and_reachability_audit.json). For the three held-out suites (Mother-1102, Tufa-Web-1102, Turducken-49, 98 games, 655 solvable levels out of 666 raw indexed levels), the Baseline column reports the frozen pre-exposure planner snapshot (32 + 10 + 30 = 72 clears, 2 wins; seeded random clears 7 + 9 + 29 = 45 levels, 1 win). In 22 community games (6 games / 26 unreachable levels in Siblings-27, 12 games / 59 unreachable levels in Clone Families-63, and 4 games / 11 unreachable levels in Mother-1102, totaling 96 unreachable levels), exhaustive state-space BFS certificates (clone_impossibility.json and clone_certificates.json) prove that an early or final map has no reachable goal state from frame 0, blocking all downstream levels in that game. Because ARCBaseGame sets GameState.WIN only when the final indexed level completes, clearing 100% of reachable levels yields 316 − 22 = 294 reachable-WIN games. At the 1,200-action budget, GiantsPlanner clears 1,793/1,794 solvable levels (99.94%) and wins 293/294* reachable games. The single remaining level (tt03 Level 3 in Siblings-27) clears in 3,299 level actions (3,471 total game actions, 3/3 WIN) under a 5,000-action budget (1,794/1,794, 100.00%). On Public-25 (1,200 steps, 12s CPU), a 4-arm component ablation (analysis/out/component_ablation_ladder.json) shows that generic (retaining only zero-prior test-time observation models and CrossLevelMemoryBank with all per-family exact solvers disabled) clears 120/183 levels (65.6%, 13/25 wins, official Scorecard $G = 58.43$ vs. 15.21/15.15 for 224/600-step open-loop probing, Wilcoxon $p = 5.96 \times 10^{-8}$, 25W/0L) vs. 42/183 (0 wins) for the coding baseline, rising to 134/183 (16/25 wins) with public decoders (no_astra) and 180/183 (24/25 wins; 183/183, 25/25 wins with confirmation rerun). On the 98 held-out games (Mother-1102, Tufa-Web-1102, Turducken-49, 655 solvable levels) evaluated with the frozen pre-exposure planner snapshot, Bare-Prior (all 8 channels C1..C8 disabled) clears 74 levels (1 win, $G = 2.93$) vs. 109 levels (6 wins, +35L [+8, +66], 11W/5L, $G = 8.22$) with OnlineInducer. Zero-prior general clears 80 levels (4 wins, $G = 5.84$ vs. 1.99 for 82L 224-step open-loop probing, $p = 4.63 \times 10^{-4}$, 42W/11L) vs. 45 (1 win) for seeded random (+35 levels, 95% CI [+8, +65], 25W/11L; +24 [+1, +49] on 90 clean-holdout games) and 72 (2 wins) for ungated full (which lost 9 levels across 2/24 misfired games burning 649 to 716 steps), while adding a 224-step certificate gate (gated) reaches 81 levels (4 wins; 41 vs. 32 on Mother-1102, +9 [0, +23]). Evaluating the unified frozen OnlineInducer across its cumulative and Leave-One-Out (LOO) operator ablation ladder rises from uncalibrated translation at 97 levels (5 wins; +17 levels over general, 95% CI [+4, +33], 10W/2L) and +Op0 novelty-gated translation at 109 levels (6 wins; +29 levels over general, 95% CI [+13, +48], 13W/0L) to +Op1 relational tile-graph (105 to 113 levels, 7 wins, 13W-15W/0L, LOO -26L [+13, +42], 14W/0L) and +Op2 switch-connectivity induction plus level-1 memory rewind (129 levels, 9 wins: 55 6W on Mother-1102, 25 2W on Tufa-Web-1102, 49 1W on Turducken-49; 125 levels 8 wins 15W/0L Op2-only; +49 levels over general, 95% CI [+24, +79], 17W/0L, LOO -26L [+4, +52], 5W/0L). Adding +Op3 visual pip-clue CSP reaches 143 to 146 levels (11 wins: 61 7W, 35 3W, 50 1W; +66 [+36, +101], 21W/0L over general, LOO -14L [+0, +36], 2W/0L, $G = 15.42$, $p = 1.07 \times 10^{-4}$ vs. general, $p = 2.93 \times 10^{-7}$ vs. open-loop, $\hat{\eta}_{\text{cert}} = 44/380 = 11.6\%$, $\hat{\eta}_{\text{game}} = 0/89 = 0.0\%$), and +Op4 1px scenery-drift mask reaches 146 levels with refit=3 (LOO -3L [+0, +8], 2W/0L) and **148 levels, 11 wins with refit=0 (+68 over general, 22W/0L, LOO -5L, $\hat{\eta}_{\text{game}} = 0/89 = 0.0\%$, +76 over 72 pre-exposure, while preserving 120/183, 13 wins, 0W/0L on Public-25)**. Stratifying all 334 unseen games across six structural mechanism families (M1 pure grid navigation $n = 22$, M2 multi-body/push/ice $n = 29$, M3 switch/door/key-lock $n = 73$, M4 pure click/toggle/flood $n = 29$, M5 visual pip-clue lattice $n = 33$, M6 coupled/dynamic multi-action $n = 148$;

mechanism_lofo_stats.json) confirms that **excluding the visual-clue family M5 entirely**, OnlineInducer lifts Held-Out 98 general ($n = 79$ non-M5 games) from 80 to **124 clears (+44.0L [+21.0, +72.0], 17W/1L,** $p = 5.3 \times 10^{-5}$**,** $\Delta G = +6.43[+2.89, +10.50]$**,** $p = 4.9 \times 10^{-4}$). On non-M5 unseen suites, OnlineInducer lifts Virgin-116 Bare-Prior +Op0+1 ($n = 115$ non-M5) by **+11.4L/seed [+0.8, +24.0] (23W/8L,** $p = 0.0080$**,** $\Delta G = +4.19$, $p = 3.2 \times 10^{-7}$) and +OI by **+12.2L/seed [+1.2, +25.0] (25W/8L,** $p = 0.0067$**,** $\Delta G = +4.07$, $p = 2.0 \times 10^{-6}$), and lifts Virgin-236 Bare-Prior +Op0+1 ($n = 222$ non-M5) by **+12.0L/seed [+1.2, +24.6] (41W/19L,** $p = 0.0118$**,** $\Delta G = +2.32$, $p = 4.5 \times 10^{-6}$), with **+14.0L [+2.0, +31.0], 8W/1L** ($p = 0.016$, $\Delta G = +1.93$) on M6 dynamic worlds alone. On the preregistered 120-game unseen suite Virgin-120 (810 levels, SHA-256 e8538024. . .), Bare-Prior + OnlineInducer reaches 99 levels (6 wins) vs. 92 (5 wins, +7L, 5W/2L, $\hat{\eta}_{\text{cert}} = 3/40 = 7.5\%$), and certificate gating lifts official Scorecard G from 4.92/5.00 (224/600-step open-loop) to **10.60 (+5.68 [+2.17, +9.50], Wilcoxon** $p = 1.83 \times 10^{-6}$**, 54W/8L)**. Across 5 external non-ARC domains rendered through a pure 64×64 uint8 pixel-grid interface (D1 MiniGrid DoorKey 250/250 clears, quadratic utility 0.869 → 0.976 vs. 0.125 under Probe64; D2 Sokoban BoxWorld 250/250 clears, quadratic utility 0.955 → 1.000 vs. 0.226 under Probe64; D3 Dungeon DiG 246 → 250/250 clears, quadratic utility 0.892 → 1.000 vs. 0.143 under Probe64; D4 Pong/Breakout 62 → 172 gated and 190/250 with OnlineInducer, +51.2 pp, $p = 2.9 \times 10^{-11}$; D5 ScienceWorld Alchemy 214 → 242 gated and 243/250 with OnlineInducer, +11.6 pp, $p = 3.8 \times 10^{-6}$; 250 tasks, 8,750 runs), 64-step probing loses −0.68 to −0.77 ($p = 1.8 \times 10^{-15}$, geometric law $R^2 = 0.997$, quadratic $R^2 = 0.9999$), while gating + OnlineInducer lifts clears from 1,022 to **1,183/1,250 (+51.2 pp Pong/Breakout** $p = 2.9 \times 10^{-11}$**, +11.6 pp Alchemy** $p = 3.8 \times 10^{-6}$**,** $\hat{\eta}_{\text{task}} = 0.0\%$).

K1b. Single-Frozen-File Bare-Prior, Leave-One-Out (LOO), and Official Scorecard G Ablations (OnlineInducer)

Suite & Ablation Arm	Active Operators	Cleared Levels (Terminal Wins)	Paired ΔL [95% Bootstrap CI], W/L	Official Scorecard G (ΔG, Paired Wilcoxon p, W/L)
Held-Out 98 (655L): Bare-Prior (0 channels)	(0,0,0,0)	74 / 655 (1 WIN)	Baseline (Mother 27, Tufa-Web 17, Turd 30)	$G = 2.93$
Held-Out 98 (655L): Bare-Prior + OnlineInducer	(1,1,1,1)	**109 / 655 (6 WIN)**	**+35L [+8, +66], 11W/5L** (Op0 vel. guard cuts losses 80% to 1)	$G = 8.22$ **(+5.28 [+1.36, +9.85])**
Held-Out 98 (655L): general (C1..C8 zero-prior)	(0,0,0,0)	80 / 655 (4 WIN)	Baseline (34 + 17 + 29; vs. 82L open-224)	$G = 5.84$ vs. 1.99 open-224 ($p = 4.63 \times 10^{-4}$, 42W/11L)
Held-Out 98 (655L): +Op1 (GraphInducer)	(1,0,0,0)	105 / 655 (7 WIN)	**+25L [+12, +41], 13W/0L** (LOO drop **-26L**, 14W/0L)	$G = 10.68$ (+4.84)
Held-Out 98 (655L): +Op1 + Op2 (ClickInducer + rewind)	(1,1,0,0)	129 / 655 (9 WIN)	**+49L [+24, +79], 17W/0L** (LOO drop **-26L**, 5W/0L)	$G = 13.71$ (+7.87)
Held-Out 98 (655L): +Op1 + Op2 + Op3 (ClueBoard CSP)	(1,1,1,0)	143 / 655 (11 WIN)	**+63L [+33, +98], 21W/0L** (LOO drop **-14L**, 2W/0L)	$G = 14.95$ (+9.11)
Held-Out 98 (655L): +Op1..Op4 (1px drift mask)	(1,1,1,1)	**148 / 655 (11 WIN)**	**+68L [+38, +103], 22W/0L** (non-M5: **+44L [+21,+72], 17W/1L,** $p = 5.3 \times 10^{-5}$)	$G = 15.42$ ($p = 1.07 \times 10^{-4}$ **vs. gen;** $p = 2.93 \times 10^{-7}$ **vs. open)**
Public-25 (183L): ungated open-224 / open-600	Open-loop	104 / 102 (11 / 10 WIN)	Baseline open-loop probing	$G = 15.21$ / 15.15
Public-25 (183L): gated general / gated + OI	Cert-gated	**120 / 183 (13 WIN)**	**+16L / +18L vs. open-loop** (0W/0L between gated + OI and gated general)	$G = 58.43$ **(+43.22 [+28.44, +58.11],** $p = 5.96 \times 10^{-8}$**, 25W/0L)**
Virgin-120 (810L): Bare-Prior vs. Bare + OI	(0..0) vs. (1..1)	**92 (5W) → 99 (6W)**	**+7L [-3, +19], 5W/2L** ($\hat{\eta}_{\text{cert}} = 3/40 = 7.5\%$, 1 step-1 abort; Op3: 0/120 games)	$G = 4.44 \rightarrow 4.87$ (+0.42)
Virgin-120 (810L): ungated open-224 vs. gated	Cert-gated	**122 (8W) → 134 (9W)**	**+12L [+3, +22], 12W/3L vs. open-224** (+42L [+17, +69], 27W/15L vs. Bare-Prior)	$G = 4.92 \rightarrow 10.60$ **(+5.68 [+2.17, +9.50],** $p = 1.83 \times 10^{-6}$**, 54W/8L)**
Virgin-116 (594L, 5 seeds): Bare-Prior → +OI (no Op3)	(0..0) → (1,1,1,0,1)	**141.8 (12.6W / 20.4W) → 152.2 (16.2W / 23.4W)**	**+10.4L/seed [+2.2, +21.4], 25W/9L** ($p = 0.0083$ **1s,** 0.017 **2s; Op3: 0/116g, LOO 0.0 [0,0]; -Op0 -10.0L** $p = 0.010$**; Op1 +60L, 12,225 steps saved)**	$G = 5.83 \rightarrow 9.57$ **(+3.74 [+2.15, +5.60],** $p = 3.5 \times 10^{-6}$**;** 5.85 → 9.52**, +3.67 [+2.11, +5.58],** $p = 1.1 \times 10^{-5}$**; LOO -Op0 -3.86** G**,** $p = 4.7 \times 10^{-7}$**; -Op1 -1.24** G**,** $p = 0.0022$)
Virgin-116 (594L, 5 seeds): Bare-Prior → +Op0+Op1	(0..0) → (1,1,0,0,0)	**141.8 (12.6W) → 151.4 (16.2W)**	**+9.6L/seed [+1.4, +20.6], 23W/9L** ($p = 0.015$**, 5/5 seeds positive)**	$G = 5.83 \rightarrow 9.69$ **(+3.86 [+2.29, +5.70],** $p = 5.6 \times 10^{-7}$)
Virgin-116 (594L, 5 seeds): open-224 vs. general + OI	Cert-gated	**146 (21W) → 243 (40W)**	**+97L vs. open-224** (235L, 39W for general; Op1: 397 certs, 15.1% aborted, 3,167 steps saved)	$G = 9.80 \rightarrow 31.35$ **(+21.55,** $p = 1.1 \times 10^{-9}$**, 62W/7L)**
Virgin-236 (1,404L, 5 seeds): Bare-Prior → +OI / +Op0+1	(0..0) → (1,1,0..1)	**235.8 (25.4W) → 246.8 (+Op0+1) / 247.4 (+OI)**	**+11.0L/seed (41W/21L,** $p = 0.013$**; non-M5 +12.0L,** $p = 0.0118$**) / +11.6L/seed [+0.4, +24.2], 44W/25L** ($p = 0.029$**; Op3: 0/236g)**	$G = 4.03 \rightarrow 6.20$ **(+2.17 [+1.25, +3.20],** $p = 2.4 \times 10^{-6}$ **for +Op0+1;** 4.05 → 6.09**, +2.04 [+1.07, +3.16],** $p = 3.0 \times 10^{-4}$ **for +OI)**
Virgin-236 (1,404L, 5 seeds): open-224 vs. general + OI	Cert-gated	**268 (29W) → 380.4 (+Op0+1) / 383 (+OI)**	**+115L vs. open-224** (general +Op0+1: +9.6L/seed, 11W/5L, LOO -Op1 -7.0L [+0.6, +15.8], $p = 0.031$)	$G = 7.36 \rightarrow 20.09$ **(+Op0+1,** $\Delta G = +0.86$ **vs. gen 19.23) /** 19.44 **(+12.08 vs. open)**
Public-25 LLM Fallback Pairing (825 runs / 250 conf.)	Cert prefix + LLM	**825 baseline + 250 confirmation**	**Causal** $L_{\text{cert}} \leq 150$**: 694W/0L/131T & 205W/0L/45T** (unguarded: 729W/64L/32T & 215W/20L/15T; post-hoc $L_{\text{cert}} \leq A$: 636W/0L/189T)	$G = 3.18 \rightarrow 58.70$ **(+55.52; 58.51 unguarded; 27.24 under** $B = \sum a$**) /** 5.42 → 59.08 **(58.92 unguarded); unseen** $q = 0\%$**: open 0.08/0.01 (0W/447L) vs. gate 3.18**

K2. Comparison with 2026 ARC-AGI-3 World-Model Preprints and Leaderboard Baselines

System / Baseline (2026)	Evaluation Scope & Score	Compute & API Regime	Unseen-Game Negative-Transfer Gate	Cross-Level Symmetry & Induction	Single-GPU Hidden LB Verified
AWM (arXiv:2610.40503)	25 public (7,448 replay / 24,604 total, 1.436x)	Cloud API + Lean	Discards 17,156 explore steps	None (per-game synthesis)	Not reported (0 LB scores)
Twin (arXiv:2608.14490)	25 public (93.3, 179/183L, 11,562 act, 0.675x)	Cloud frontier API	Replay validation of twin	None (LLM code twin)	Not reported (0 LB scores)
Schema (arXiv:2609.39140)	25 public (99.2, 25/25W) + MazeBench	Cloud frontier API	Prompted rule revision	None (LLM rule prompt)	Not reported (0 LB scores)
Memento 3 (arXiv:2610.11794)	25 public (100.0, 7,518 act, 0.439x) + Atari	Cloud frontier API	Reflective text rulebook	None (text rulebook)	Not reported (0 LB scores)
Self-Harness (arXiv:2610.30499)	25 public split (57.0% vs. 54.0% base)	Cloud frontier API	Holdout CI [-1.44, +7.35]	None (harness code)	Not reported (0 LB scores)
VISTA (arXiv:2610.02200)	25 public (100.0, 7,302 act, 30.7M tok/g)	Cloud multimodal API	Visual harness memory	None	Not reported (0 LB scores)
Kepler (arXiv:2610.00834)	25 public games (auditable world model)	Cloud frontier API	1-step conditional interrupt	None	Not reported (0 LB scores)
State Re-Obs (arXiv:2610.26632)	25 public games (re-observation diagnostic)	Cloud frontier API	None (diagnostic study)	None	Not reported (0 LB scores)
Workspace Opt (arXiv:2605.09650)	25 public games (36.0% -> 38.4%)	Cloud frontier API	Prompt/workspace evolution	None	Not reported (0 LB scores)
Prob. Induction (arXiv:2609.01815)	Grid/program induction suite	Cloud API + SMC	Bayesian posterior filter	Library wake-sleep	Not reported (0 LB scores)
Yi-Chia Chen (#1 LB, Topic 755795)	110 hidden (75.56 LB, 16-26% > human act)	1 GPU INT3/NVFP4 SGLang	Stage-switched agentic RL	None (pure LLM+RL)	Yes (75.56 LB; no dilution theorem)
JKU Linz / Sestak (#7 LB Writeup)	110 hidden (54.52 LB, 5-6h cliff stall)	1 GPU REAP-320 MoE 4-bit KV	Cliff context compaction	None (pure LLM)	Yes (54.52 LB; reports 5-6h wall)
Zatravkin (#6) & xz (#4, Topic 755202)	110 hidden (54.0 LB, throughput plateau)	1 GPU open-weights	Heuristic prompt/action loop	None	Yes (54.0 LB; reports plateau)
pompom555 (#15 LB Writeup)	110 hidden (46.20 LB)	1 GPU open-weights	Token-budget heuristic	None	Yes (46.20 LB; no symbolic gate)
GPT-6 Astra / Google Adapters [3]	25 public (99.9% / 97.5%)	$18,817-$26,098/run API	Unconstrained cloud retries	None	Ineligible for offline Kaggle GPU
Official Human Baseline [7]	25 public (183/183L, 17,135 act, 1.000x)	486 human players (342 runs)	Human visual prior	284/597 click & 120/597 prefix	Reference baseline (b_l)
GiantsPlanner (Ours)	**316 entries (G = 99.58 raw 7,401 / G = 100.00 loop-erased 7,070, 0.413x) + 250 non-ARC**	**Single offline GPU ($0 API, 0 tokens on Public-25)**	**One-way certificate $C(F_0)$ + auto_dead_level + $L_{\text{cert}} \leq 150$ (694W/0L)**	**Multi-scale D_4 L2 (79.6% recall) + 4-op OnlineInducer (+68L, 22W/0L)**	**Yes (30 runs, $k = 0$ 24.83±2.12, peak 28.38, gated 25.70 vs. 20.76/21.08, $U = 0$, $p = 0.0083$)**

Table K2. Head-to-head comparison against all 16 contemporary 2026 world-model preprints, top Hidden-110 leaderboard solutions, cloud frontier harnesses, and the official human baseline. On Public-25 in 0 LLM tokens ($0 API vs. VISTA's 30.7M tokens/game), GiantsPlanner's raw online trajectory takes 7,401 actions ($G = 99.58$, 0.432x human baseline 17,135, beating AWM's 24,604 total online actions by 3.32x and Memento 3's 7,518 online actions), while stripping within-level state-space cycles yields a 7,070-action real-engine replay (0.413x human, 0 resets, 169/183 levels at the 115% cap vs. AWM's 7,448-action replay, Memento 3's 7,518, and VISTA's 7,302 actions) that achieves $G = 100.00$ across all 25/25 games (lifting cd82 from 93.96 to 100.00, lp85 from 98.08 to 100.00, and sp80 from 97.54 to 100.00).

References

Numbered as in the main paper.